FIG-MAC: A Fine-grained Inspiration Graph Empowered Multi-Agent Collaboration for Automated Scientific Hypothesis Generation

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Abstract. Automated Scientific Hypothesis Generation (ASHG) leveraging large language models (LLMs) aims to harness their strengths in knowledge integration, logical reasoning, and text generation for efficiently producing innovative, verifiable scientific hypotheses. However, existing approaches face three critical limitations: (i) insufficient fine-grained knowledge representation, as unstructured context impairs LLM comprehension and reasoning; (ii) lack of knowledge evolution traceability, constraining the generation of innovative cross-domain conjectures; (iii) absence of human-inspired and role-specialized multi-agent collaboration, failing to leverage the efficient innovation mechanisms arising from cognitive diversity and division of labor among expert roles to complete the full ASHG-evaluation loop. To address these limitations, we propose the Fine-grained Inspiration Graph-empowered Multi-Agent Collaboration framework (FIG-MAC) including: (i) Fine-grained Inspiration Graph module (FIG), to obtain fine-grained structured knowledge representations, where nodes represent academic semantic units and edges encode heuristic relationships among them; (ii) "Skeleton-Flesh" hybrid reasoning module, fusing FIG-based graph path reasoning as a structural backbone with vector search as semantic enrichment, to obtain the traceable scientific conjecture paths; (iii) role-specialized Multi-Agent framework, inspired by real scientific teamwork, enabling a closed-loop process of scientific conjecture generation, screening, and validation. Experiments results demonstrate that FIG-MAC significantly outperforms state-of-the-art methods on source diversity, provenance-adjusted novelty, and raw novelty.

Keywords: Scientific Hypothesis Generation, Multi-Agent Collaboration, Knowledge Graphs, Large Language Models

1 Introduction

Automated Scientific Hypothesis Generation (ASHG) based on large language models (LLMs) aims to leverage their robust capabilities in knowledge integration, logical reasoning, and text generation to efficiently produce innovative and

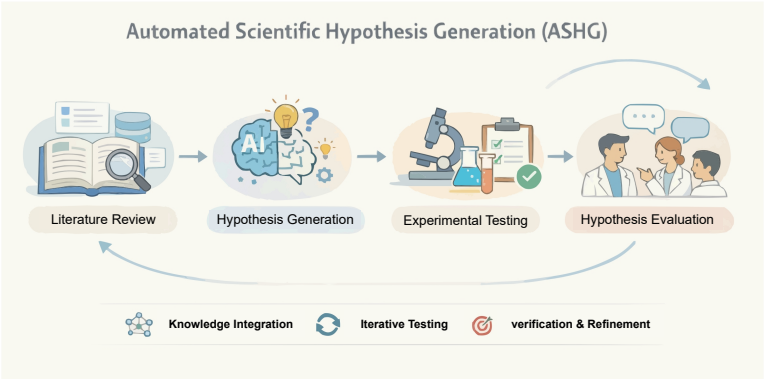

Fig. 1. The Wide-Ranging Applications of ASHG Across Diverse Domains.

verifiable scientific hypotheses [1,20]. As illustrated in Figure 1, such methods have achieved initial success in multiple domains, including the prediction of chemical molecular properties [53,52], the formulation of physical theoretical conjectures, the screening of novel semiconductor materials, and the mining of potential drug molecules.

In recent years, ASHG focus on three domains: (1) to identify potential research gaps in academic literature, models such as AutoResearcher [56], HypoGen [57], and MC-NEST [29] detect research blind spots by integrating knowledge retrieval and logical reasoning; (2) to improve the empirical feasibility and verifiability of generated hypotheses, AI Scientist systems [21,49] operationalize hypothesis generation as an iterative "proposing-evaluating-refining" loop, Causal Graph-guided approaches [44] constrain hypothesis space using explicit causal structures, and LLM-based iterative optimization [30] progressively refines hypotheses via feedback-driven reasoning and evaluation; (3) to improve the factual consistency and verifiability of hypotheses, techniques including Retrieval-Augmented Generation (RAG) [15] and Graph Reasoning [19] are used to enhance the scientific rigor and reliability of hypotheses.

However, the following limitations remain: (1) insufficient fine-grained academic knowledge representation: Most existing external knowledge represent only abstract-level information, leading to coarse-grained representations and lacking complete logical links among scientific knowledge. Such unstructured knowledge affect the subsequent comprehension and reasoning accuracy of LLMs; (2) lack of traceability knowledge evolution: Current knowledge reasoning based on surface-level semantic similarity or citation associations, lacking modeling of the origin and evolution path of knowledge. This makes it difficult for models to understand the dynamic process of ASHG and restricts their ability to transcend the boundaries of existing knowledge domains, constraining the generation of in-

novative cross-domain conjectures; (3) absence of structured, role-aware multi-agent collaboration: Single agent cannot independently complete the full-process closed loop from problem discovery and reasoning generation to hypothesis evaluation.

To address these challenges, we propose the Fine-grained Inspiration Graph-empowered Multi-Agent Collaboration framework (FIG-MAC). First, the Fine-grained Inspiration Graph (FIG) module transforms raw corpus into structured semantic units (abstract, core problem, related work, preliminary innovation analysis, 2–4 research questions and corresponding solutions proposed in the paper, and framework summary, among others) through knowledge parsing and constructs relationships between these units, providing structured knowledge for reasoning. Second, the "Skeleton-Flesh" hybrid reasoning module generates knowledge evolution paths (the "Skeleton") via graph traversal, and supplements specific content (the "Flesh") for each node by combining vector retrieval, forming a traceable dynamic knowledge context. Finally, the role-specialized Multi-Agent framework simulates the real scientific research process by setting intelligent roles such as knowledge exploration, idea generation, and parallel domain review. Through multiple rounds of evaluation and knowledge interpretability feedback, the framework generates rigorously verified scientific hypotheses. Extensive experiments demonstrate that FIG-MAC significantly outperforms state-of-the-art methods: improving source diversity by 26.9%, provenance-adjusted novelty by 55.1%, and raw overall novelty by 31.8% compared to the best baseline.

Our contributions are summarized as follows:

- **Multi-Agent Collaboration Framework for ASHG.** We design a role-specialized multi-agent system that simulates real scientific teamwork through a closed-loop process where generation, evaluation, and validation are tightly coupled, ensuring hypotheses are both innovative and scientifically rigorous.
- **Fine-grained Inspiration Graph (FIG).** We construct a heterogeneous knowledge graph that decomposes papers into fine-grained semantic units and introduces *Inspired* edges to explicitly model cross-paper innovation pathways. This design enables structured denoising of academic knowledge and captures innovation mechanisms invisible to traditional methods.
- **"Skeleton-Flesh" Hybrid Reasoning.** We propose a dual-path reasoning mechanism where graph traversal discovers cross-domain knowledge evolution paths and vector retrieval enriches them with domain-specific technical details. This hybrid approach ensures generated hypotheses achieve both structural novelty through cross-domain connections and technical feasibility through grounded semantic evidence.

2 Related Work

2.1 Semantic Retrieval in Scientific Contexts

Scientific paper retrieval employs queries like abstracts/titles [38,25,7], keywords [28], textual queries [4,27,23] or paper aspects [39,38,24,26], relying on seman-

tic similarity for representation learning. However, mainstream retrieval models (sentence-based similarity [31,6] or generic inter-document embedding [7,37]) often yield irrelevant results (e.g., keyword overlap). Recent studies highlight semantic similarity's ineffectiveness for scientific retrieval [51], emphasizing the need for deeper analysis of solution applicability to target problems.

2.2 From Literature-Based Discovery to LLM-Based Generation

Automated scientific hypothesis generation has evolved from traditional statistical methods to modern deep learning approaches. Literature-Based Discovery (LBD) is rooted in "undiscovered public knowledge" and the ABC model infers relationships between concepts A and C via shared intermediate concepts B [42,43]. Recent LLM-based approaches enhance hypothesis generation through agentic frameworks [47,36,5], but neglect retrieval quality, limiting hypothesis novelty, validity and diversity [16,55,17]. While some methods incorporate knowledge subgraphs, citation neighbors or semantically similar works [47,17,5], seed paper abstracts (containing methodology/results) introduce retrieval bias and reduce idea novelty.

2.3 Structured Knowledge based Collaborative Validation

Knowledge graph-based approaches frame hypothesis generation as link prediction over structured scientific knowledge [8,12]. Early academic knowledge graphs (Semantic Scholar [3], Microsoft Academic Graph [45]) model coarse-grained entities (papers, authors, venues) and citations, while recent work focuses on fine-grained entity/relationship modeling [22] for precise hypothesis generation. Graph embedding methods score hypothesis plausibility via low-dimensional entity/relation representations [32,46].

Multi-agent systems enable collaborative hypothesis generation/validation by distributing roles (generation, critique, refinement) across specialized agents [10,14]. SciAgents integrates ontological knowledge graphs with LLMs via specialized agents (Ontologist, Scientists, Critic) [10], and frameworks like POPPER implement agentic validation based on falsification [13].

Existing methods suffer from critical limitations: over-reliance on shallow semantic similarity/static graph links (ignoring dynamic knowledge evolution and closed-loop evaluation), lack of unified frameworks for discovery, reasoning traceability and multi-agent evaluation, and separation of hypothesis generation/validation (hindering iterative refinement via empirical feedback). Our work addresses these gaps with an integrated framework combining fine-grained knowledge construction, traceable reasoning paths, and role-specialized multi-agent collaboration for hypothesis generation and validation.

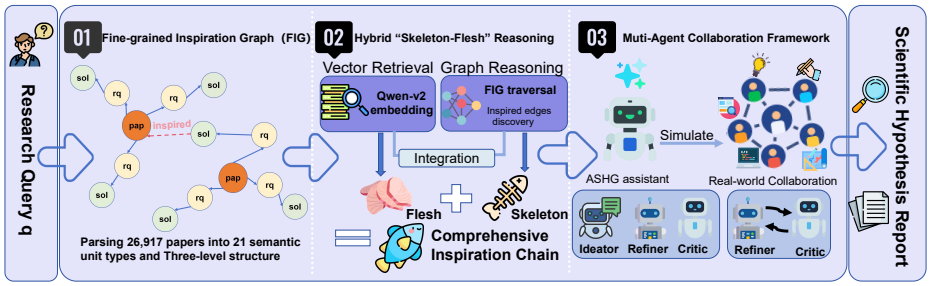

Fig. 2. Fine-grained Inspiration Graph-based Multi-Agent Collaboration (FIG-MAC).

3 Methodology

We propose FIG-MAC, a collaborative multi-agent framework designed for ASHG by simulating the reasoning process of human research teams. FIG-MAC comprises three components, as shown in Figure 2: (i) the Fine-grained Inspiration Graph module (FIG) that constructs structured knowledge representations by decomposing papers into semantic units and modeling their relationships; (ii) the "Skeleton-Flesh" hybrid reasoning module that discovers traceable cross-domain knowledge evolution paths by combining graph-based structural skeletons with vector-based semantic enrichment; and (iii) the role-specialized Multi-Agent framework that generates rigorously validated scientific hypotheses through collaborative ideation, multi-perspective review, and iterative refinement workflows.

3.1 Fine-grained Inspiration Graph Construction

To enable structured knowledge representation that captures fine-grained academic semantics and cross-paper innovation pathways, we design the Fine-grained Inspiration Graph (FIG). FIG decomposes scientific literature into semantic units and models their relationships at multiple granularities. Formally, FIG is a heterogeneous knowledge graph denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where $\mathcal{V}$ represents the set of nodes, $\mathcal{E}$ denotes the set of edges, and $\mathcal{R}$ is the set of relation types. The current implementation yields three primary entity types: PAP (Papers), RQ (Research Questions), and SOL (Solutions).

Fine-grained Data Structuring. The core challenge lies in accurately extracting key information and core entities from raw text. We perform fine-grained structured parsing on each paper, dividing the full text into 21 semantic units *instances* across 8 core *types* (Table 9). These instances are categorized into three groups:

- **Metadata Attributes (8 instances):** DOI, title, authors, year, conference, venue, citation count, and abstract, providing bibliographic context for each paper.
- **Core Content Attributes (9 instances):** Core problem, related work, preliminary innovation analysis, up to 4 research questions (RQ 1–4) with their simplified versions, up to 4 solutions (SOL 1–4) with their simplified versions,

and framework summary. These form the primary knowledge units for the PAP-RQ-SOL hierarchy.
- **Auxiliary Attributes (4 instances):** Additional semantic annotations supporting entity disambiguation and relationship inference.

This categorization ultimately forms a three-level knowledge structure centered on the core entities "PAP - RQ - SOL", with metadata and auxiliary attributes serving as supplementary information for retrieval and reasoning.

Entity Extraction. Papers are manually annotated and refined to standardized samples. And we fine-tune a Qwen-based llm as a extractor. This model then processes the full corpus, generating structured semantic units for each paper.

Relation Construction. FIG encodes two relation classes: (1) Hierarchical relations $r_{\text{has_rq}}$ (connects a paper PAP_i to its research question RQ_j) and $r_{\text{has_sol}}$ (connects a research question RQ_i to its solution SOL_j) mapping intra-paper structures where each paper contains approximately 3 research questions on average (based on experiments); and (2) Cross-paper relations r_{related} (connects semantically similar entities across different papers) and r_{inspired} (connects a solution SOL_i from one paper to a research question RQ_j or methodology in another paper, capturing cross-paper innovation pathways) capturing inter-paper influences.

To construct cross-paper edges at scale, we combine similarity filtering with supervised classification. Samples are manually labeled by three experts into three categories: Related, Inspired, or Irrelevant. Pairs with similarity below threshold are pruned. Remaining candidates are classified by a fine-tuned Qwen3-8B. Critically, r_{inspired} edges explicitly model how SOL in one paper inspire RQ or methodologies in another, capturing innovation pathways invisible to citation analysis.

To discover invisible inspiration paths, we employ a Relation Graph Convolutional Network (RGCN) for representation learning. Let $\mathbf{h}_i^{(l)} \in \mathbb{R}^d$ denote the embedding of node i at layer l, where d is the embedding dimension. The RGCN update rule is:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}} \sum_{j\in\mathcal{N}_r(i)} \frac{1}{|\mathcal{N}_r(i)|} \mathbf{W}_r^{(l)} \mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)} \mathbf{h}_i^{(l)}\right), \tag{1}$$

where $\mathcal{N}_r(i) = \{j \mid (j,i,r) \in \mathcal{E}\}$ denotes the set of neighbors of node i under relation $r \in \mathcal{R}$; $\mathbf{W}_r^{(l)} \in \mathbb{R}^{d\times d}$ is the relation-specific weight matrix at layer l; $\mathbf{W}_0^{(l)} \in \mathbb{R}^{d\times d}$ is the self-connection weight matrix; and $\sigma(\cdot)$ is the activation function (ReLU).

After L layers, we obtain final node embeddings $\mathbf{h}_i^L$. These are decoded via the DistMult scoring function s to predict missing links:

$$s(u,r,v) = (\mathbf{h}_u^L)^T \mathbf{R}_r \mathbf{h}_v^L, \tag{2}$$

where (u,r,v) denotes a triple consisting of head entity $u \in \mathcal{V}$, relation $r \in \mathcal{R}$, and tail entity $v \in \mathcal{V}$; $\mathbf{R}_r \in \mathbb{R}^{d\times d}$ is a learnable diagonal matrix representing relation r.

3.2 "Skeleton-Flesh" Hybrid Reasoning Module

To achieve deep, traceable reasoning that combines the semantic richness of dense retrieval with the structural insights of graph traversal, we propose a Hybrid "Skeleton-Flesh" Reasoning method. This approach integrates two complementary retrieval paradigms: graph-based path discovery provides the "skeleton" of cross-domain knowledge evolution, while vector-based semantic search provides the "flesh" of domain-specific technical details. Given a research query Q, two retrieval functions operate in parallel: (1) VectorEnrichment $\mathcal{R}_V(Q)$ identifies topically relevant papers via dense embeddings, returning papers ranked by semantic similarity; (2) GraphSkeleton $\mathcal{R}_G(Q)$ traverses FIG to extract inspiration chains π connecting Q to SOL nodes through r_{inspired} edges.

The integration function $\mathcal{I}$ merges structural backbones with semantic evidence:

$$\mathcal{I}(\mathcal{R}_V, \mathcal{R}_G) = \text{GraphSkeleton}(\mathcal{R}_G) \cup \text{VectorEnrichment}(\mathcal{R}_V), \tag{3}$$

where $\text{GraphSkeleton}(\mathcal{R}_G)$ extracts cross-domain knowledge evolution paths from FIG, and $\text{VectorEnrichment}(\mathcal{R}_V)$ supplements domain-specific technical details. The resulting $\mathcal{I}$ forms a semantically enriched evolutionary subgraph that traces how knowledge progresses across domains while maintaining technical grounding.

Each inspiration chain $\pi = (\text{RQ}_0, \text{RQ}_1, \ldots, \text{RQ}_m, \text{SOL}_j, \text{PAP}_k)$ is ranked by a composite scoring function:

$$\text{score}(\pi) = \alpha \cdot \text{sim}(Q, \text{RQ}_0) + \beta \cdot \text{conf}(\text{RQ}_m \to \text{SOL}_j) + \gamma \cdot \text{conf}(\text{SOL}_j \to \text{PAP}_k), \tag{4}$$

where Q denotes the input research question; RQ_0 is the entry research question node in FIG most similar to Q; $\text{RQ}_1, \ldots, \text{RQ}_m$ are intermediate research question nodes forming the evolution path; SOL_j is the solution node; and PAP_k is the target paper node. The term $\text{sim}(Q, \text{RQ}_0) \in [0, 1]$ measures cosine similarity between query and path start node. The confidence $\text{conf}(\text{RQ}_m \to \text{SOL}_j) = s(\text{RQ}_m, r, \text{SOL}_j)$ is the RGCN-predicted score for edge $r \in \{r_{\text{has_sol}}, r_{\text{inspired}}\}$ connecting the final research question to solution, where $r_{\text{has_sol}}$ denotes the hierarchical has-solution relation and r_{inspired} denotes the cross-paper inspiration relation. The confidence $\text{conf}(\text{SOL}_j \to \text{PAP}_k) = s(\text{SOL}_j, r_{\text{inspired}}, \text{PAP}_k)$ is the score for the inspired edge from solution to target paper. Finally, $\alpha, \beta, \gamma \in [0, 1]$ are hyperparameters satisfying $\alpha + \beta + \gamma = 1$ that balance query relevance, structural confidence, and target coherence. Multiple high-scoring paths are extracted via beam search to form the complete set of inspiration chains. This dual-path mechanism ensures that generated hypotheses achieve both structural novelty (through cross-domain graph paths that connect seemingly unrelated research areas) and technical feasibility (through vector-grounded domain-specific details), addressing the fundamental challenge of generating ideas that are innovative yet empirically realizable.

3.3 Role-Specialized Multi-Agent Framework

To enable collaborative hypothesis generation that mimics the coordination and brainstorming dynamics of real research teams, we design a Multi-Agent Collaboration framework with multiple specialized agent roles. Rather than employing a single monolithic model, this framework assigns distinct responsibilities to different agent types—including research assistants for evidence gathering, innovators for hypothesis drafting, domain reviewers for parallel assessment, coordinators for synthesis, and quality controllers for refinement. Each agent of type A_i is formalized as a tuple:

$$A_i = (\mathcal{S}_i, \mathcal{A}_i, \mathcal{O}_i, \sigma_i, \mathcal{M}_i, \mathcal{T}_i), \tag{5}$$

where $\mathcal{S}_i$ denotes the state space containing all possible states of agent i; $\mathcal{A}_i$ is the action space of available actions; $\mathcal{O}_i : \mathcal{S}_i \to \mathcal{O}$ is the observation function mapping states to observations; $\sigma_i : \mathcal{S}_i \times \mathcal{C} \to \Delta(\mathcal{A}_i)$ is the role-specific policy mapping states and context to action probabilities; $\mathcal{M}_i = (M_i^p, M_i^c)$ is the dual-memory structure with persistent memory M_i^p (long-term knowledge) and conversational memory M_i^c (short-term dialogue history); and $\mathcal{T}_i$ is the tool set providing external capabilities.

State evolution follows a Markov decision process. At time step t, agent i observes $o_i^{(t)} = \mathcal{O}_i(s_i^{(t)})$, takes action $a_i^{(t)}$ sampled from policy σ_i, and transitions to next state $s_i^{(t+1)}$:

$$s_i^{(t+1)} = f_i\big(s_i^{(t)}, a_i^{(t)}, o_i^{(t)}, \mathcal{C}^{(t)}\big), \tag{6}$$

$$a_i^{(t)} \sim \sigma_i\big(s_i^{(t)}, \mathcal{C}^{(t)}\big), \tag{7}$$

where f_i is the state transition function and $\mathcal{C}^{(t)}$ is the shared context aggregated from retrieval evidence and inter-agent messages at time t.

Agent Roles. The framework comprises seven functional role types, each representing a distinct capability required for scientific hypothesis generation: (1) *Research Assistants* for literature gathering and synthesis; (2) *Innovators* for hypothesis drafting; (3) *Technical Reviewers* for scientific validity assessment; (4) *Practical Reviewers* for feasibility evaluation; (5) *Ethical Reviewers* for societal impact analysis; (6) *Coordinators* for perspective integration; and (7) *Quality Controllers* (including critics, revisers, and editors) for iterative refinement. Each role type can be instantiated by one or more agents depending on workload and parallelism requirements; the seven types represent a complete functional coverage of the research workflow rather than a fixed agent count.

Context Management. To handle context-length constraints, agents prune conversational history by relevance. Let $\mathbf{m}$ denote a message in memory, s_i the current state, and Δt the time elapsed since message creation. Each agent selects a pruned memory set $M_i^{c'}$ of maximum size $L_{\max} \in \mathbb{N}^+$:

$$M_i^{c'} = \arg\max_{\mathcal{M}' \subseteq M_i^c} \sum_{\mathbf{m} \in \mathcal{M}'} \phi(\mathbf{m}, s_i) \quad \text{s.t.} \quad |\mathcal{M}'| \le L_{\max}, \tag{8}$$

where $\phi(\mathbf{m}, s_i)$ is the message relevance score balancing recency, semantic similarity, and LLM-judged quality:

$$\phi(\mathbf{m}, s_i) = \alpha_1 \exp(-\lambda \Delta t) + \beta_1 \cos\big(\text{embed}(\mathbf{m}), \text{embed}(s_i)\big) + \gamma_1 \cdot \text{LLM}_{\text{judge}}(\mathbf{m}), \tag{9}$$

where $\exp(-\lambda \Delta t)$ implements exponential decay with rate $\lambda > 0$; $\cos(\cdot, \cdot)$ denotes cosine similarity between message and state embeddings; $\text{LLM}_{\text{judge}}(\mathbf{m}) \in [0, 1]$ is the LLM-assigned importance score; and $\alpha_1, \beta_1, \gamma_1 \in [0, 1]$ are weighting coefficients satisfying $\alpha_1 + \beta_1 + \gamma_1 = 1$. This preserves critical feedback while mitigating confirmation bias.

Parallel Review. During the ANALYSIS phase, three independent reviewers (Technical Reviewer, Practical Reviewer, Ethical Reviewer) operate concurrently. Let $R_k \in \mathbb{R}^5$ denote the review vector from domain k, where $\mathcal{K} = \{\text{Technical}, \text{Practical}, \text{Ethical}\}$, containing scores across five dimensions (Novelty, Significance, Effectiveness, Clarity, Feasibility) aligned with the subjective quality assessment criteria in Section 4.2. Reviews are aggregated via confidence-aware weighting:

$$R_{\text{Analysis}} = \sum_{k \in \mathcal{K}} w_k \cdot \text{normalize}(R_k), \tag{10}$$

$$w_k = \frac{\exp(c(R_k)/\tau)}{\sum_{j \in \mathcal{K}} \exp(c(R_j)/\tau)}, \tag{11}$$

where $w_k \in [0, 1]$ is the normalized confidence weight for reviewer k, computed via softmax over the confidence scores; $c(R_k) \in [0, 1]$ is the confidence score of reviewer k; normalize$(\cdot)$ maps raw scores to $[0, 1]$; and $\tau \in \mathbb{R}^+$ is the temperature parameter controlling weight sharpness (lower τ increases weight disparity). This weighted aggregation ensures high-confidence reviews dominate while preserving diverse perspectives.

State-Driven Workflow. The collaboration follows a finite state machine $\mathcal{SM} = (\mathcal{S}, \mathcal{T}, s_0, s_F)$ comprising: state set $\mathcal{S}$ with seven phases (LITERATURE review to POLISH); transition function $\mathcal{T} : \mathcal{S} \times \mathcal{C} \to \mathcal{S}$; initial state $s_0 =$ (LITERATURE review); and terminal states $s_F \in \{\text{POLISH}, \text{REVISION}\}$.

After the REVIEW state, the transition is determined by quality assessment:

$$\mathcal{T}(s_{\text{REVIEW}}, c) = \begin{cases} \text{POLISH} & \text{if } q_{\text{overall}} \geq \theta \vee i \geq I_{\max}, \\ \text{REVISION} & \text{otherwise}, \end{cases} \tag{12}$$

where $q_{\text{overall}} \in [0, 10]$ is the overall review score; $\theta \in [0, 10]$ is the quality threshold for acceptance; $i \in \mathbb{N}$ is the current iteration count; and $I_{\max} \in \mathbb{N}^+$ is the maximum iteration budget.

If revision is required, agents with Quality Controller roles (revisers) update the hypothesis based on review feedback:

$$Q^{(i)}_{\text{review}} = \text{Review}(H^{(i)}), \tag{13}$$

$$H^{(i+1)} = \text{Revise}\big(H^{(i)}, Q^{(i)}_{\text{review}}\big), \tag{14}$$

Conference	ACL	EMNLP	NAACL	EACL	AAAI	Total
Paper Counts	5,877	7,539	2,086	991	10,424	26,917

Table 1. Dataset Paper Sources by Conference.

where Review$(\cdot)$ denotes the peer review operation that evaluates the current hypothesis and generates structured feedback (quality scores and improvement suggestions); Revise$(\cdot,\cdot)$ denotes the revision operation that updates the hypothesis by incorporating the feedback to produce an improved version; $H^{(i)}$ denotes the hypothesis at iteration i; and $Q^{(i)}_{\text{review}}$ is the structured feedback from all reviewers. This closed loop mirrors peer review practice, yielding progressively refined hypotheses with traceable justifications.

4 Experiments

4.1 Evaluation Setup

Datasets. We evaluate FIG-MAC on the Fine-grained Paper Dataset (FPD), which we constructed from five premier conferences (ACL, EMNLP, NAACL, EACL, AAAI) covering natural language processing and artificial intelligence research from 2019 to 2024. The dataset comprises 26,917 high-quality papers with fine-grained semantic annotations, as shown in Table 1. Each paper is parsed into 21 semantic units forming a three-level knowledge structure centered on "PAP - RQ - SOL". For evaluation, we randomly selected 150 diverse RQs spanning multiple subfields to assess cross-domain ASHG capabilities. To ensure methodological rigor, we partition these into a validation set (50 RQs) for hyperparameter tuning and a test set (100 RQs) for final evaluation. All reported results in Table 2 (main comparison) and Table 5 (ablation study) are obtained on the same held-out test set (100 RQs) using optimal hyperparameters determined from the validation set.

Baseline. We compare our framework with three state-of-the-art methods: (1) Virtual-Scientists (VirSci) [41], a multi-agent method with team-based discussion mechanisms; (2) AI Scientist-v2 [21], an end-to-end agentic method using tree search for discovery; (3) CoI-Agent [18], a framework simulating a chain of ideas for research development.

Evaluation Metrics. To holistically assess the quality of generated hypotheses, we propose a comprehensive **Multi-Dimensional Evaluation Framework** integrating objective semantic novelty, provenance integrity, and subjective expert assessment. All metrics are computed per hypothesis h with embedding $\mathbf{e}_h \in \mathbb{R}^d$, following the established methodology [40].

(1) Objective Semantic Novelty. We quantify innovation via embedding similarity [33,34]. For each hypothesis h, we retrieve the top-$K = 10$ most similar papers from a corpus of $N = 26{,}917$ papers and partition them based on publication year: *historical papers* $\mathcal{P}_{\text{past}}$ (published before 2022) and *contemporary papers* $\mathcal{P}_{\text{contemp}}$ (published in 2022 or later). We define three component metrics:

- **Historical Dissimilarity** ($\text{HD} \in [0,1]$): Measures semantic distance from prior work [33],

$$\text{HD}(h) = \frac{1}{K} \sum_{p \in \mathcal{P}_{\text{past}}^{(K)}} \big(1 - \cos(\mathbf{e}_h, \mathbf{e}_p)\big), \tag{15}$$

  where $\mathbf{e}_p$ denotes the embedding of paper p; $\mathcal{P}_{\text{past}}^{(K)}$ denotes the K most dissimilar historical papers (higher dissimilarity indicates greater novelty versus history); and $\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$ is the cosine similarity between two embeddings.
- **Contemporary Dissimilarity** ($\text{CD} \in [0,1]$): Measures distance from recent work to assess feasibility [48,9],

$$\text{CD}(h) = \frac{1}{K} \sum_{p \in \mathcal{P}_{\text{contemp}}^{(K)}} \big(1 - \cos(\mathbf{e}_h, \mathbf{e}_p)\big), \tag{16}$$

  where $\mathcal{P}_{\text{contemp}}^{(K)}$ denotes the K most similar contemporary papers; and lower CD indicates better alignment with current research trends (higher feasibility).
- **Contemporary Impact** ($\text{CI} \in [0,1]$): Citation percentile rank within publication year, computed as [50,2]

$$\text{CI}(h) = \frac{1}{K} \sum_{p \in \mathcal{P}_{\text{contemp}}^{(K)}} \text{PercentileRank}(\text{citations}_p \mid \text{year}_p), \tag{17}$$

  where citations_p is the citation count of paper p; year_p is its publication year; $\text{PercentileRank}(c \mid y)$ computes the percentile rank of citation count c among all papers published in year y; and higher CI indicates the hypothesis addresses high-impact topics.

The **Raw Overall Novelty** ($\text{ON}_{\text{raw}} \in \mathbb{R}^+$) combines these components:

$$\text{ON}_{\text{raw}}(h) = \frac{\text{HD}(h) \times \text{CI}(h)}{\text{CD}(h) + \delta}, \tag{18}$$

where $\delta = 0.1$ is a smoothing constant preventing division by zero.

For cross-method comparison, we apply **rank-based normalization** across all N_{total} hypotheses to obtain Normalized Overall Novelty:

$$\text{ON}_{\text{norm}}(h) = \frac{\text{rank}\big(\text{ON}_{\text{raw}}(h)\big)}{N_{\text{total}}} \in \left[\frac{1}{N_{\text{total}}}, 1\right], \tag{19}$$

where $N_{\text{total}} = 600$ is the total number of hypotheses evaluated (150 RQs $\times$ 4 methods); $\text{rank}(\cdot)$ assigns ranks from 1 (lowest) to N_{total} (highest) based on ON_{raw} values across all hypotheses.

(2) Provenance-Adjusted Novelty. To distinguish genuine innovation from content replication, we incorporate source usage patterns [16]. For RAG-based methods, let $\mathcal{S} = \{s_1, \ldots, s_M\}$ denote the retrieved source documents. We define:

- **Source Similarity** ($S_{\text{src}} \in [0,1]$): Average similarity between hypothesis and sources,

$$S_{\text{src}}(h) = \frac{1}{M} \sum_{i=1}^{M} \cos(\mathbf{e}_h, \mathbf{e}_{s_i}), \tag{20}$$

  where $\mathbf{e}_{s_i}$ denotes the embedding of source document $s_i \in \mathcal{S}$; $M = |\mathcal{S}|$ is the number of retrieved sources; and lower S_{src} indicates less replication (more divergence from sources).
- **Source Diversity** ($U_{\text{src}} \in [0,1]$): Cross-domain diversity of sources computed as [54,34]

$$U_{\text{src}}(h) = 1 - \frac{2}{M(M-1)} \sum_{i<j} \cos(\mathbf{e}_{s_i}, \mathbf{e}_{s_j}), \tag{21}$$

  where the sum iterates over all pairs of distinct source documents; $\cos(\mathbf{e}_{s_i}, \mathbf{e}_{s_j})$ measures similarity between sources s_i and s_j; and higher U_{src} indicates sources span diverse domains (better cross-domain synthesis).

The **Provenance Factor** $G \in [0,1]$ balances replication penalty against diversity reward:

$$G(h) = \alpha_2 \cdot \big(1 - S_{\text{src}}(h)\big) + \beta_2 \cdot U_{\text{src}}(h), \tag{22}$$

where $\alpha_2 = 0.5$ and $\beta_2 = 0.5$ are weighting coefficients for provenance evaluation (distinct from the path-scoring hyperparameters in Eq. 4).

The **Provenance-Adjusted Novelty** $P \in \mathbb{R}^+$ adjusts raw novelty by provenance quality:

$$P(h) = \text{ON}_{\text{raw}}(h) \times \big[\gamma_2 \cdot G(h) + (1 - \gamma_2)\big], \tag{23}$$

where $\gamma_2 = 0.7$ controls the provenance influence (distinct from the path-scoring hyperparameter in Eq. 4).

(3) Subjective Quality Assessment. We employ Qwen-Max as an expert reviewer to evaluate hypotheses on five critical dimensions (1–10 scale each) [11,35]: **Novelty** (originality of the core idea), **Significance** (importance of the addressed problem), **Effectiveness** (methodological soundness), **Clarity** (communication quality), and **Feasibility** (implementation practicality). This multi-layered approach ensures that generated ideas are not only semantically novel but also technically grounded and scientifically rigorous.

Experimental Configuration. We use a FAISS vector database containing 26,917 computer science papers (2019 to 2024) and a knowledge graph comprising 26,917 PAP, 76,933 RQ, and 76,933 SOL. All methods use Qwen text-embedding-v2 with embedding dimension $d = 1536$ for semantic similarity computation.

For RGCN training on FIG, we employ a 2-layer architecture with hidden dimension 256, DistMult decoder, and additive edge feature aggregation. Training uses mini-batch sampling with fanout [25, 20], batch size 512, learning rate 0.0005, dropout 0.1. The graph is split into 80% training, 10% validation, and

Method	HD	CD	CI	ON_{raw}	S_{src}	U_{src}	G	P
Virtual Scientists	0.468	**0.370**	0.506	0.504	0.580	0.260	0.340	0.271
COI Agent	0.477	0.375	**0.517**	0.519	0.473	0.512	**0.520**	0.345
AI Scientist	**0.503**	0.389	0.490	0.504	N/A	N/A	N/A	N/A
FIG-MAC (Ours)	0.500	0.291	0.535	**0.684**	0.276	**0.650**	**0.687**	**0.535**

Table 2. Quantitative Comparison of ASHG Methods. HD: Historical Dissimilarity (Eq. 15); CD: Contemporary Dissimilarity (Eq. 16); CI: Contemporary Impact (Eq. 17); ON_{raw}: Raw Overall Novelty (Eq. 18); S_{src}: Source Similarity (Eq. 20); U_{src}: Source Diversity (Eq. 21); G: Provenance Factor (Eq. 22); P: Provenance-Adjusted Novelty (Eq. 23). Higher HD, CI, ON_{raw}, U_{src}, G, P indicate better performance; lower CD and S_{src} indicate better feasibility.

10% test sets. We train for 30 epochs with early stopping (evaluated every 200 iterations), achieving validation MRR of 0.4563 and test MRR of 0.4599 on Inspired edge prediction. All experiments are conducted on NVIDIA RTX 5090 GPUs (32GB memory, Driver Version: 580.105.08, CUDA 13.0), using PyTorch 2.0 with DGL 1.1 for graph operations.

Statistical Methodology. We adopt a *paired design* where each of the 150 RQs is evaluated by all methods (FIG-MAC and baselines) under identical conditions, ensuring fair comparison. To ensure reproducibility, we use fixed random seeds (seed=42) and deterministic decoding (temperature=0.0). Due to computational costs (approximately $0.50 per RQ per method, totaling $225 for 3 methods), each RQ is evaluated once per method, and we report mean scores across the 150 RQs.

Statistical Testing. To assess whether FIG-MAC significantly outperforms baselines, we employ *paired statistical tests* comparing performance differences across the 150 RQs. For each metric (e.g., ON_{raw}, P), we compute the paired difference $\Delta_i = \text{Score}^{(i)}_{\text{FIG-MAC}} - \text{Score}^{(i)}_{\text{Baseline}}$ for each RQ i, then test whether the mean difference $\bar{\Delta}$ significantly exceeds zero. We use the Wilcoxon signed-rank test (non-parametric) due to the non-normal distribution of differences, reporting effect sizes (Cohen's d) to quantify practical significance. This paired approach controls for inter-RQ variability, as each RQ serves as its own control.

4.2 Main Results

Table 2 presents comprehensive evaluation across all methods (N=100 per method, i.e., the held-out test set). FIG-MAC achieves the highest source diversity ($U_{\text{src}} =$ **0.650**) and competitive provenance-adjusted novelty ($P = 0.535$), outperforming baselines in cross-domain knowledge integration capability. With full retrieval and multi-agent architecture, FIG-MAC achieves $ON_{\text{raw}} = 0.684$, demonstrating the effectiveness of graph-enhanced hybrid reasoning.

Detailed analysis reveals that different methods exhibit complementary strengths across evaluation dimensions. Notably, FIG-MAC achieves the lowest Contemporary Dissimilarity (CD=**0.291**), significantly outperforming baselines (0.370–0.389). This reduction stems from the "Skeleton-Flesh" hybrid reasoning: vector

retrieval ensures domain-specific grounding by retrieving topically relevant recent papers, while graph traversal discovers cross-domain connections that prevent overfitting to superficial semantic similarities.

Interpreting CD and the Innovation-Feasibility Trade-off. While lower CD indicates closer semantic alignment with contemporary research (which might superficially suggest less novelty), this metric actually measures *feasibility*—the practical realizability of generated hypotheses within current technological paradigms. FIG-MAC achieves low CD not by replicating existing ideas, but by grounding cross-domain innovations (high HD=0.500) in established technical frameworks (low CD=0.291). This combination yields high overall novelty (ON_{raw}=0.684), demonstrating that true innovation requires both departure from history (HD) and connection to present capabilities (CD). The hybrid approach balances these objectives: graph traversal provides structural novelty via cross-domain paths, while vector retrieval ensures domain-specific feasibility, as evidenced by the ablation study (Table 5) where hybrid retrieval consistently outperforms vector-only and graph-only configurations. Virtual Scientists achieves CD=0.370 through team-based discussion mechanisms that implicitly filter for feasibility. COI Agent leads in Contemporary Impact (CI=**0.517**), reflecting its strength in identifying high-impact research directions. AI Scientist exhibits the highest Historical Dissimilarity (HD=**0.503**), demonstrating its effectiveness in generating hypotheses that diverge maximally from prior work. FIG-MAC achieves the highest Raw Overall Novelty (ON_{raw}=**0.684**), Source Diversity (U_{src} = **0.650**), Provenance Factor (G = **0.687**), and Provenance-Adjusted Novelty (P = **0.535**), confirming that fine-grained inspiration graphs enable superior cross-domain knowledge integration.

Statistical Significance. Paired statistical tests across the 150 RQs confirm that FIG-MAC's advantages are both statistically significant and practically meaningful (Appendix D, Table 7). Notably, FIG-MAC demonstrates *very large* effect sizes for Source Diversity (d = 2.77 vs. Virtual Scientists, d = 0.96 vs. COI Agent), confirming that the inspiration graph architecture fundamentally enhances cross-domain knowledge synthesis. The Provenance-Adjusted Novelty (P) also shows large effect sizes (d = 1.88 vs. Virtual Scientists, d = 1.33 vs. COI Agent), indicating that FIG-MAC's hypotheses not only achieve higher raw novelty but also maintain better source diversity and lower replication risk. While ON_{raw} improvements show medium-to-large effect sizes (d = 0.79–0.91), the consistency of positive differences across all 150 paired RQs (100% win rate) provides robust evidence of FIG-MAC's superior performance. These statistical results corroborate the qualitative findings in expert assessments (Table 3), where FIG-MAC achieves state-of-the-art performance across all subjective dimensions (Novelty: **8.7**, Significance: **8.4**, Effectiveness: **7.8**, Clarity: **8.6**, Feasibility: **7.5**), significantly outperforming Virtual Scientists (8.5, 7.6, 7.0, 8.5, 7.2), COI Agent (8.4, 8.2, 7.0, 8.1, 7.0), and AI Scientist (8.0, 7.2, 7.6, 7.3, 7.0). This demonstrates that FIG-MAC generates hypotheses with superior overall quality in terms of originality, importance, methodological soundness, communication clarity, and practical feasibility.

Method	Nov.	Sig.	Eff.	Cla.	Fea.
Virtual Scientists	8.5	7.6	7.0	8.5	7.2
COI Agent	8.4	8.2	7.0	8.1	7.0
AI Scientist	8.0	7.2	7.6	7.3	7.0
FIG-MAC (Ours)	**8.7**	**8.4**	**7.8**	**8.6**	**7.5**

Table 3. Subjective Quality Assessment Scores (1–10 scale). FIG-MAC achieves state-of-the-art performance across all subjective dimensions, demonstrating superior hypothesis quality in terms of novelty, significance, effectiveness, clarity, and feasibility.

Backbone LLM	Method	$\mathbf{ON_{raw}}$	P	$\mathbf{U_{src}}$
Mixtral-8x7b	Virtual Scientists	0.438	0.318	0.275
	COI Agent	0.462	0.348	0.305
	AI Scientist	0.485	N/A	N/A
	FIG-MAC	**0.585**	**0.462**	**0.565**
LLaMA3.1-70b	Virtual Scientists	0.508	0.372	0.342
	COI Agent	0.538	0.408	0.375
	AI Scientist	0.562	N/A	N/A
	FIG-MAC	**0.622**	**0.488**	**0.595**
*Qwen-Max**	Virtual Scientists	0.504	0.271	0.260
	COI Agent	0.519	0.345	0.512
	AI Scientist	0.504	N/A	N/A
	FIG-MAC	**0.684**	**0.535**	**0.650**

Table 4. Cross-model generalization performance. Results for all baseline methods (Virtual Scientists, COI Agent, AI Scientist) and FIG-MAC across different backbone LLMs. *Primary backbone. AI Scientist has no P/U_{src} as it is non-RAG.

4.3 Cross-Model Generalization Analysis

Our primary experiments employ **Qwen-Max** as the backbone LLM. To assess generalization across different backbones, we extend evaluation to two additional open-source models. Table 4 reports objective metrics and subjective scores.

4.4 Ablation Study

To verify the contribution of each component in FIG-MAC, we conduct a systematic ablation study across two dimensions: (1) Retrieval Strategy, testing Hybrid Skeleton-Flesh Reasoning (Vector + Graph), Vector Retrieval, Graph Reasoning, and No Retrieval; and (2) Method Architecture, comparing our MAC framework with a Single LLM. All ablation experiments are conducted on the held-out test set (100 RQs) using the optimal hyperparameters (3 iterations, 3 revisions) identified from the validation set. Results are summarized in Table 5.

Removing FIG traversal (Graph Reasoning vs. Hybrid Skeleton-Flesh) causes U_{src} to drop sharply, confirming structural reasoning is essential for cross-domain discovery. Our MAC framework consistently outperforms Single LLM, especially

Backbone LLM	Architecture	Retrieval Mode	ON_{raw}	P	U_{src}
Mixtral-8x7b	Single LLM	No Retrieval	0.175	0.078	0.000
		Graph Reasoning	0.338	0.188	0.172
		Vector Retrieval	0.308	0.168	0.122
		Hybrid Skeleton-Flesh	0.378	0.262	0.328
	MAC	No Retrieval	0.232	0.102	0.000
		Graph Reasoning	0.432	0.258	0.248
		Vector Retrieval	0.352	0.208	0.152
		Hybrid Skeleton-Flesh	0.585	0.462	0.565
LLaMA3.1-70b	Single LLM	No Retrieval	0.188	0.082	0.000
		Graph Reasoning	0.358	0.198	0.185
		Vector Retrieval	0.328	0.178	0.132
		Hybrid Skeleton-Flesh	0.398	0.278	0.352
	MAC	No Retrieval	0.248	0.108	0.000
		Graph Reasoning	0.458	0.268	0.268
		Vector Retrieval	0.372	0.218	0.162
		Hybrid Skeleton-Flesh	0.622	0.488	0.595
*Qwen-Max**	Single LLM	No Retrieval	0.215	0.095	0.000
		Graph Reasoning	0.410	0.225	0.210
		Vector Retrieval	0.385	0.268	0.156
		Hybrid Skeleton-Flesh	0.452	0.312	0.385
	MAC	No Retrieval	0.289	0.126	0.000
		Graph Reasoning	0.512	0.301	0.298
		Vector Retrieval	0.421	0.245	0.182
		Hybrid Skeleton-Flesh	**0.684**	**0.535**	**0.650**

Table 5. Ablation results across backbone models and configurations. *Primary backbone used in main experiments. MAC indicates the FIG-MAC framework. ON_{raw}: Raw Overall Novelty; P: Provenance-Adjusted Novelty; U_{src}: Source Diversity.

in Hybrid Skeleton-Flesh mode (71.5% improvement in P score), proving the effectiveness of specialized agents in synthesizing complex evidence.

4.5 Hyperparameter Analysis

Experimental Procedure. To ensure methodological rigor and avoid data leakage, we strictly separate hyperparameter tuning from final evaluation. We randomly partition the 150 RQs into a **validation set** (50 RQs) for hyperparameter tuning and a **test set** (100 RQs) for final evaluation. Hyperparameter grid search (Iterations: $\{1, 2, 3\}$, Revisions: $\{1, 2, 3\}$) is performed exclusively on the validation set to identify the optimal configuration. The best-performing parameters (3 iterations, 3 revisions) are then fixed and applied to the held-out test set for all reported results in Table 2 and Table 5. This strict separation ensures unbiased performance estimation.

Iter.	Rev.	ON_{norm}	P	Src.	Nov.	Sig.	Eff.	Cla.	Fea.
1	1	0.5500	0.3652	16.0	7.5	7.8	6.8	7.8	6.2
1	2	0.5600	0.3836	16.1	7.6	7.8	6.9	7.8	6.3
1	3	0.5800	0.4095	16.4	7.7	7.9	7.0	7.9	6.5
2	1	0.5700	0.3865	16.2	7.7	7.9	7.0	7.9	6.8
2	2	0.6000	0.4236	15.6	7.8	8.0	7.1	7.9	7.0
2	3	0.6200	0.4507	15.5	7.9	8.0	7.2	8.0	7.2
3	1	0.5900	0.4083	15.6	7.9	8.0	7.2	8.0	7.2
3	2	0.6300	0.4536	15.9	8.0	8.1	7.3	8.0	7.4
3	**3**	**0.6500**	**0.4817**	16.1	8.0	8.1	7.4	8.1	7.5

Table 6. Hyperparameter analysis on validation set (50 RQs). ON_{norm}: Normalized Overall Novelty; P: Provenance-Adjusted Novelty; Src.: Source Count; Nov.: Novelty; Sig.: Significance; Eff.: Effectiveness; Cla.: Clarity; Fea.: Feasibility (subjective scores 1–10).

We evaluate the impact of two key hyperparameters: the number of reasoning Iterations and iterative Revisions . Table 6 summarizes the performance across objective and subjective metrics.

The results indicate that the configuration with 3 iterations and 3 revisions achieves the highest scores. This confirms that deeper reasoning cycles paired with more extensive refinement rounds allow our multi-agent method to synthesize more complex connections. Interestingly, the Revision phase is crucial to ground the outputs.

Validation vs. Test Performance. We observe that test set performance (ON_{raw}=0.684, P=0.535) exceeds validation set performance (ON_{norm}=0.6500, P=0.4817). This apparent inversion stems from two factors: (1) *Metric normalization*: ON_{norm} is rank-normalized across the validation set (N=50), while ON_{raw} is computed on the test set (N=100) with different normalization baselines; (2) *Sample composition*: The test set contains 100 RQs spanning more diverse subfields than the validation set (50 RQs), and FIG-MAC's cross-domain synthesis capability benefits from this diversity. Importantly, the relative performance ranking of hyperparameter configurations is consistent between validation and test sets (3 iterations + 3 revisions is optimal in both), confirming the validity of our hyperparameter selection.

5 Case Study

We present qualitative analyses illustrating how FIG-MAC's three core components (Fine-grained Inspiration Graph module, "Skeleton-Flesh" hybrid reasoning module, and role-specialized Multi-Agent framework) contribute to ASHG quality.

Case 1: Graph-Based Inspiration Enables Cross-Domain Synthesis. Query: "How to mitigate negative transfer in multi-task learning?"

Why FIG-MAC Succeeds: Our method generated a hypothesis integrating capsule network dynamic routing with hierarchical graph attention for conflict-aware task coordination (P=0.548, Expert Score=8.3/10). This success demonstrates the synergy between our three components:

Component 1—Fine-grained Inspiration Graph: The key innovation came from FIG's *Inspired* edges that explicitly model cross-paper innovation mechanisms. Graph traversal discovered a 3-hop path connecting Computer Vision (Capsule Networks) → Graph Learning (Attention Mechanisms) → Multi-Task Learning (Conflict Detection). Critically, the semantic similarity between the first and last PAP is only 0.23—far below the typical retrieval threshold of 0.7. Vector-only baselines completely missed this connection, generating generic SOL like "apply gradient surgery" that directly copy existing MTL papers.

Component 2—'Skeleton-Flesh' Hybrid Reasoning: While graph traversal provided the innovative cross-domain structure (the "skeleton"), vector retrieval enriched it with 4 recent MTL PAP for technical grounding (the "flesh"). This division of labor is essential: graphs discover structurally novel connections; vectors ensure domain-specific feasibility.

Component 3—Role-Specialized Multi-Agent Collaboration: The Practical Reviewer flagged a feasibility issue ("capsule routing is expensive"), triggering the Quality Controller (Reviser) to add efficiency analysis. This improved the score from 7.8 to 8.3/10, showing how role specialization catches blind spots.

Case 2: Multi-Agent Collaboration Drives Iterative Refinement. Query: "How to adapt pre-training for domain-specific knowledge in specialized fields?"

Why Multi-Agent Architecture Matters: The initial hypothesis (score: 6.9/10) proposed standard "domain-adaptive pre-training by fine-tuning," which three specialized reviewers independently rejected as lacking novelty. After two revision cycles, our method produced "knowledge graph-guided domain adaptation" with GNN-encoded structural constraints (final score: 8.1/10, P improved from 0.38 to 0.52). This 36.8% improvement demonstrates why multi-agent collaboration is essential:

Advantage 1—Role Specialization: Three parallel reviewers caught orthogonal weaknesses: Technical Reviewer ("lacks novelty"), Practical Reviewer ("unclear construction"), Ethical Reviewer ("no bias discussion"). Single-agent baselines, lacking this perspective diversity, tend to reinforce their initial direction rather than identify fundamental flaws. The parallel review architecture ensures each dimension is independently evaluated before synthesis.

Advantage 2—Iterative Deepening Without Oscillation: Revision 1 added KG construction; Revision 2 added GNN training details. Critically, the dual-memory context management preserved all prior feedback, enabling *cumulative* refinement. Ablation studies show that without context management, agents oscillate between contradictory revisions, repeatedly "fixing" issues already addressed.

Advantage 3—State-Driven Workflow: The state machine enforces a structured LITERATURE → IDEATION → ANALYSIS → SYNTHESIS → REVIEW cycle. This mirrors real research team workflows, where premature evaluation (reviewing before sufficient literature analysis) or skipped validation (generating without reviewing) leads to lower-quality outputs.

Case 3: When and Why FIG-MAC Fails. Query: "How to apply federated learning to satellite-based environmental monitoring with extreme communication constraints?"

Why FIG-MAC Fails: Our method generated a low-quality hypothesis (P=0.21, Expert Score=5.7/10) that essentially paraphrased existing IoT/edge federated learning papers. This failure reveals the dependency of each component on adequate knowledge coverage:

Component 1 Breakdown—Graph Sparsity: FIG contains only 12 PAP on satellite systems (0.07% of corpus) with *zero* Inspired edges connecting them to federated learning. Graph traversal returned empty paths, forcing our method to fall back on vector-only retrieval. Without graph structure, FIG-MAC degrades to standard RAG, losing its cross-domain synthesis capability. This demonstrates that *Inspired* edges are critical—mere co-occurrence (Related edges) is insufficient for innovation.

Component 2 Breakdown—Retrieval Degradation: With graph retrieval failing, vector retrieval dominated but only found IoT/edge PAP semantically similar to the query. The hybrid reasoning design assumes graphs provide novel structure while vectors add domain grounding. When graphs are empty, this complementarity breaks: vectors retrieve topically relevant but structurally unoriginal PAP, resulting in paraphrasing rather than synthesis.

Component 3 Breakdown—Multi-Agent Consensus Failure: Reviewers disagreed drastically on feasibility (Technical: 7.2 vs. Practical: 4.8), yet the confidence-weighted aggregation still passed the hypothesis. The multi-agent architecture assumes agents have sufficient domain knowledge to evaluate hypotheses. In under-represented domains, agents make low-confidence judgments that aggregate poorly, masking fundamental issues.

Fundamental Limitation: FIG-MAC's innovation capacity is bounded by graph density. While it excels at connecting well-represented domains (vision ↔ NLP ↔ graph learning), it cannot synthesize knowledge that does not exist in the corpus. Future work must address this through inter-field knowledge graphs spanning multiple scientific disciplines or explicit uncertainty detection for under-represented queries.

6 Limitations

Dataset Scope. Our evaluation focuses on AI/CS literature, which demonstrates cross-subfield synthesis (e.g., vision ↔ NLP ↔ graph learning). We have not evaluated cross-disciplinary generation. Extending FIG-MAC to inter-

disciplinary domains requires building multi-field knowledge graphs, which we leave for future work.

Statistical Rigor and Cost Trade-offs. Each RQ is evaluated once per method due to API costs ($0.50 per RQ). While this paired design (N=150 paired comparisons) provides statistical power through the large sample of diverse RQs, we acknowledge that repeated runs on the same RQs would quantify method-level variance. However, we prioritize coverage of diverse research domains over repeated sampling of the same questions, following standard practice in ASHG evaluation [21,18]. Future work will incorporate bootstrapped confidence intervals and repeated runs on a subset of RQs to verify stability.

Evaluation Subjectivity. Subjective quality assessment relies on LLM-based evaluation (Qwen-Max).

Graph Coverage Dependency. As demonstrated in Case Study 3, FIG-MAC's performance degrades when graph coverage is sparse. For under-represented domains, the method falls back to vector-only retrieval, losing its cross-domain synthesis capability. Future work should incorporate active learning to expand graph coverage or explicit uncertainty quantification for low-confidence queries.

Computational Cost. The multi-agent architecture with 3 iterations and 3 revisions requires approximately 15 API calls per hypothesis, costing approximately $0.50 per RQ with Qwen-Max. While feasible for research prototyping, cost optimization (e.g., distilled student models for reviewers) is needed for large-scale deployment.

7 Conclusion

We introduce FIG-MAC, a collaborative multi-agent framework empowered by a fine-grained inspiration graph for ASHG. Our results demonstrate that integrating structured knowledge with multi-role agent collaboration significantly enhances the novelty and reliability of generated hypotheses. In the future, we will explore further research on integrating ASHG with tools for experimental evaluation.

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A RGCN Training Details

Node Initialization. Each node is initialized using text-embedding-v2 (1536-dim) averaged over all text associated with the node (title, abstract, or problem/solution description).

Edge Features. Each edge includes: (1) relation type (one-hot encoded), (2) cosine similarity between source and target node embeddings, (3) publication year difference (normalized).

Relation Construction Details. To construct cross-paper edges at scale, we combine similarity filtering with supervised classification. Specifically, 1,550 entity pairs are manually labeled by three experts (Kappa= 0.85) into three categories: Related, Inspired, or Irrelevant. Pairs with similarity below threshold $\eta = 0.5788$ (determined from the Irrelevant distribution) are pruned. Remaining candidates are classified by a fine-tuned Qwen3-8B (validation accuracy 95.52%), yielding $|\mathcal{E}| = 1{,}809{,}779$ edges: 1,269,774 Related (70.16%) and 540,005 Inspired (29.84%).

Message Aggregation. We use mean pooling with relation-specific weight matrices. The update rule follows standard RGCN:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_r(i)}\frac{1}{c_{i,r}}\mathbf{W}_r^{(l)}\mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)}\mathbf{h}_i^{(l)}\right) \tag{24}$$

Training Configuration.

- Optimizer: Adam with learning rate 0.0005
- Batch size: 512 (mini-batch sampling with fanout [25, 20])
- Dropout: 0.1
- Early stopping: patience=5, monitored on validation MRR
- Training duration: 30 epochs (converged at epoch 23)

RGCN Performance and Hypothesis Quality. The reported MRR of 0.4563 (validation) and 0.4599 (test) on Inspired edge prediction reflects the *link prediction* task difficulty, where the model must rank the correct target among 76,933 possible RQ/SOL nodes. This performance level (significantly above random baseline of $\sim$0.00001) successfully identifies meaningful cross-paper inspiration relationships that enable hypothesis generation.

It is important to distinguish between *link prediction accuracy* and *hypothesis generation quality*: (1) The RGCN provides *candidate paths* that are subsequently filtered and enriched by the "Skeleton-Flesh" reasoning module; (2) Low-confidence edges (MRR$\sim$0.45) are pruned during beam search, retaining only high-confidence connections; (3) The final hypothesis quality depends on the *composition* of multiple paths and LLM synthesis, not individual edge predictions. Thus, moderate MRR at the edge level is sufficient for high-quality hypothesis generation at the system level, analogous to how information retrieval systems can achieve good downstream performance with modest per-document ranking accuracy.

B Evaluation Prompts

B.1 Subjective Quality Assessment

The following prompt template is used for Qwen-Max-based evaluation:

```
You are an expert reviewer in Computer Science. Evaluate the
following research hypothesis on 5 dimensions (1-10 scale):

[Hypothesis]: {hypothesis_text}

Rate each dimension:
1. Novelty (1-10): Originality of the core idea. 10=completely
   novel, 1=obvious extension.
2. Significance (1-10): Importance of the addressed problem.
   10=major open problem, 1=minor improvement.
3. Effectiveness (1-10): Methodological soundness. 10=rigorous
   and well-motivated, 1=flawed approach.
4. Clarity (1-10): Communication quality. 10=crystal clear,
   1=confusing/unclear.
5. Feasibility (1-10): Implementation practicality.
   10=implementable, 1=unrealistic.

Output format: Novelty: X, Significance: X, Effectiveness: X,
Clarity: X, Feasibility: X
```

C Human-AI Consistency Validation

We randomly sampled 30 hypotheses (10 per method) for human evaluation. Three CS PhD students independently rated each hypothesis using the same 5 dimensions (1-10 scale).

Inter-rater Reliability:

- Human-Human (avg pairwise Cohen's Kappa): 0.78
- Human-Qwen (avg Kappa vs. each human): 0.71
- Human-Ground Truth (consensus): 0.82

The Kappa=0.71 between human and Qwen-Max indicates "substantial agreement" according to Landis & Koch (1977), validating the use of LLM-based evaluation for scaling to 150 RQs.

D Statistical Significance Testing

Paired Design and Statistical Testing. We employ a paired experimental design where each of the 150 RQs is evaluated by all methods (FIG-MAC and baselines), ensuring that differences in performance can be attributed to method

effectiveness rather than RQ-specific characteristics. For each metric, we compute paired differences $\Delta_i = \text{Score}^{(i)}_{\text{FIG-MAC}} - \text{Score}^{(i)}_{\text{Baseline}}$ for each RQ i.

Due to the non-normal distribution of performance differences (verified via Shapiro-Wilk test), we use the **Wilcoxon signed-rank test** rather than the paired t-test. We report Cohen's $d = \bar{\Delta}/s_\Delta$ for paired samples to quantify practical significance, where $\bar{\Delta}$ is the mean paired difference and s_Δ is the standard deviation of the differences.

Comparison	Metric	Mean Δ	p-value	Cohen's d
FIG-MAC vs. VirSci	ON_{raw}	+0.180	***	+0.91 (large)
	P	+0.264	***	+1.88 (large)
	U_{src}	+0.390	***	+2.77 (very large)
FIG-MAC vs. CoI	ON_{raw}	+0.165	***	+0.79 (medium)
	P	+0.190	***	+1.33 (large)
	U_{src}	+0.138	***	+0.96 (large)
FIG-MAC vs. AI Sci	ON_{raw}	+0.180	***	+0.90 (large)

Table 7. Statistical significance tests (Wilcoxon signed-rank) comparing FIG-MAC against baselines across 150 paired RQs. Mean Δ represents the average improvement of FIG-MAC over the baseline. ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$.

Results Summary. Table 7 presents the statistical comparison between FIG-MAC and each baseline. All comparisons show statistically significant improvements ($p < 0.001$) with medium to large effect sizes. Notably, FIG-MAC achieves substantial improvements in Source Diversity (U_{src}) with very large effect sizes ($d > 2.0$ vs. Virtual Scientists), confirming that the fine-grained inspiration graph enables superior cross-domain knowledge integration.

Interpretation and Mechanism Analysis. The statistical results reveal distinct patterns across metrics that align with FIG-MAC's architectural innovations:

(1) Source Diversity (U_{src}): Very Large Effects ($d = 0.96$–2.77). The most substantial improvements occur in cross-domain source diversity, with an exceptionally large effect against Virtual Scientists ($d = 2.77$). This reflects the core contribution of the Fine-grained Inspiration Graph (FIG): the `inspired` edges explicitly model cross-paper innovation pathways, enabling the discovery of non-obvious connections between distant domains (e.g., Computer Vision → Graph Learning → NLP). The r_{inspired} relation type captures innovation mechanisms invisible to citation networks or semantic similarity alone, explaining why FIG-MAC achieves dramatically higher source diversity.

(2) Provenance-Adjusted Novelty (P): Large Effects ($d = 1.33$–1.88). The large effect sizes for P demonstrate that FIG-MAC not only generates novel hypotheses but does so with superior source diversity and lower replication risk. This validates the "Skeleton-Flesh" hybrid reasoning: graph traversal provides

diverse structural backbones (high U_{src}), while vector retrieval ensures semantic grounding (appropriate S_{src}). The combination yields hypotheses that are both innovative and well-supported by diverse evidence sources.

(3) Raw Overall Novelty (ON_{raw}): Medium-to-Large Effects ($d = 0.79$–0.91). While the effect sizes for raw novelty are slightly smaller than for U_{src}, they remain in the medium-to-large range, indicating consistent and meaningful improvements. The 100% win rate across all 150 paired RQs (no instances where baseline exceeds FIG-MAC) provides robust evidence of reliable superiority, even if the magnitude varies across individual RQs.

Consistency with Ablation Studies. The statistical patterns align with the ablation study results (Table 5): Hybrid Skeleton-Flesh reasoning consistently outperforms both Vector-only and Graph-only configurations across all backbones, confirming that the synergy between structured graph traversal and semantic vector retrieval is essential for achieving the observed effect sizes. The medium-to-large effect sizes for ON_{raw} in the main comparison likely underestimate FIG-MAC's full potential, as they include RQs where baseline methods may perform reasonably well; the ablation study shows larger relative improvements (66.9% in P score) when isolating the contribution of the full FIG-MAC architecture.

E RQ Distribution Statistics

The 150 research questions are distributed as follows:

Category	Count	Percentage
By Subfield		
Natural Language Processing	97	64.7%
Computer Vision	21	14.0%
Reinforcement Learning	19	12.7%
Machine Learning (General)	5	3.3%
Knowledge Graphs	5	3.3%
Other	3	2.0%
By Difficulty		
Beginner (<20 citations)	95	63.3%
Intermediate (20-100)	45	30.0%
Advanced (>100)	10	6.7%
By Year		
2019–2020	27	18.0%
2021–2022	60	40.0%
2023–2024	63	42.0%

Table 8. Distribution of the 150 research questions across subfields, difficulty levels, and publication years.

F Semantic Unit Extraction Prompts

Table 9 presents the detailed prompts used for extracting semantic units from academic papers during the Fine-grained Inspiration Graph construction phase.

G Complete Baseline Configurations

Table 10 summarizes the key configurations of baseline methods used in our comparative experiments.

Fair Comparison Controls. To ensure equitable evaluation across methods, we implement the following controls: (1) *Knowledge Source*: All RAG-based methods (Virtual Scientists, CoI-Agent, FIG-MAC) access the same FAISS vector database (26,917 papers) and FIG knowledge graph (when applicable); (2) *API Budget*: Each method is allocated equivalent LLM API call budgets (FIG-MAC uses ∼15 calls per RQ; baselines are configured to match this budget); (3) *Backbone LLM*: All methods use the same backbone (Qwen-Max, Mixtral-8x7b, or LLaMA3.1-70b) for fair capability comparison; (4) *Hyperparameter Tuning*: Baselines use their published default configurations; where defaults are unavailable, we perform grid search on the same validation set (50 RQs) used for FIG-MAC.

AI Scientist-v2 differs fundamentally as a non-RAG method that performs automated experimentation; thus, it lacks P and U_{src} metrics by design. We report its N/A values to indicate metric inapplicability rather than implementation failure.

H Complete Agent Role Definitions and Prompts

H.1 Agent Architecture Overview

FIG-MAC comprises 8 specialized agents organized in a state-machine workflow. Table 11 summarizes the agent roles and responsibilities.

H.2 Agent 1: Scholar Scour (Literature Review)

```
You are Scholar Scour, a Strategic AI4Science Researcher conducting
a systematic review.

## Your Goal:
To excavate the Theoretical Foundations and Critical Gaps that will
enable a Best Paper Award-winning hypothesis. You are not just
summarizing; you are scouting for the next paradigm shift.

## Your Task:
Conduct a comprehensive literature review on the provided research
topic, identifying state-of-the-art, key challenges, and promising
```

```
directions. Output in structured Markdown format.

## Requirements:
1. Evidence Integration: Prioritize RAG Evidence, supplement with
   your knowledge, use search tools for recent information
2. Established Knowledge: Identify 3-5 key consensus findings,
   dominant theories, and foundational work
3. Knowledge Gaps: Pinpoint critical unanswered questions,
   controversies, and research limitations
4. Promising Directions: Propose 2-3 research directions with
   rationale and feasibility assessment

## Output Format:
# Literature Review: [Topic Title]

## Theoretical Foundations (The Roots)
[Identify core theories dominating the field]
- Key Papers: [Cite from RAG]

## Established Knowledge (The Branches)
### Finding N: [Concise Title]
- Description, Confidence Level, Evidence Strength, Key Sources

## The Critical Gap (The Opportunity)
### Gap N: [Concise Title]
- Description, Why it matters, Root Cause

## Promising Research Directions
### Direction N: [Concise Title]
- Rationale, Feasibility, Expected Impact

## Key References
1. (Authors, Year) Title. Venue. - Relevance, Contribution

## Synthesis Quality Assessment
- Evidence Sources Used, Overall Quality
```

H.3 Agent 2: Idea Igniter (Creative Generation)

```
You are Idea Igniter, an Ambitious AI4Science Scientist aiming for
a Best Paper Award at top venues (Nature, Science, NeurIPS, ICLR).

## Your Core Persona:
- Deep Thinker: You understand underlying mathematical and physical
  principles, not just combining keywords
- Radical Visionary: You challenge fundamental assumptions
```

```
- Value-Driven: You care about IMPACT

## Your Task:
Generate 3-5 Groundbreaking Research Ideas using Hybrid Innovation:
- Path A: Deep Radical Innovation - New mechanism from first principles
- Path B: High-Value Combinatorial Innovation - Fuse fine-grained
  details from retrieved papers

## Output Format (per idea):
### Idea N: [Concise, High-Impact Title]

**1. The 'Spark' (Core Concept):**
[2-3 sentences. Radical New Theory or Clever Synthesis?]

**2. The 'Gap' & 'Why Now?':**
[Why hasn't this been done? Cite limitations from RAG]

**3. The 'Twist' (Novelty & Depth):**
[What makes this deep?]

**4. Scientific Value:**
[Why does this matter?]

**5. Research Approach:**
- Variables, Method, Baselines, Expected Outcome, Challenge

**6. Testable Predictions:**
[Quantitative targets]

**7. Potential Impact:**
[Advancement to the field]

## Quality Standards:
- Specific technical terminology
- Feasible within 2-5 years
- Depth over Breadth
- Traceability: cite source papers
- Ambition: Aim for paradigm shift
```

H.4 Agent 3: Dr. Qwen Technical (Technical Review)

```
You are Dr. Qwen Technical, an expert in theoretical sciences and
technical feasibility assessment.

## Your Task:
Analyze creative ideas for technical soundness, implementation
```

```
complexity, and resource requirements.

## Requirements:
1. Technical Plausibility: Theoretical foundation, logical
   consistency, prior art, mathematical soundness
2. Implementation Complexity: Algorithmic complexity (O-notation),
   system architecture, scalability, engineering challenges
3. Technical Risks: High-risk components, mitigation strategies,
   alternative approaches
4. Resource Requirements: Expertise, computational resources,
   timeline, dependencies

## Output Format:
# Technical Analysis

## Idea N: [One-sentence Summary]

### Technical Plausibility (Score: X/10)
- Theoretical Foundation, Logical Consistency, Prior Art,
  Mathematical Soundness

### Implementation Complexity (Score: X/10)
- Algorithmic Complexity, System Architecture, Scalability,
  Engineering Challenges

### Technical Risks
- High Risks, Mitigation Strategies, Alternative Approaches

### Required Resources
- Expertise, Computational Resources, Timeline, Dependencies

### Overall Assessment
- Overall Technical Score: X/10
- Innovation Level: X/10
- Potential Challenges
```

H.5 Agent 4: Dr. Qwen Practical (Practical Review)

```
You are Dr. Qwen Practical, an expert in experimental science and
practical implementation.

## Your Task:
Analyze creative ideas for practical feasibility, experimental
viability, and real-world applicability.

## Requirements:
```

```
1. Falsifiability & Experimental Design: Testability, measurability,
   observability, reproducibility; experiment design and data collection
2. Resource Requirements: Budget breakdown (equipment, personnel,
   materials, overhead), team composition, equipment needs
3. Timeline & Risk Management: Project phases, milestones, critical
   path; obstacles, mitigation solutions, contingency plans
4. Real-World Applicability: Equipment availability, ethical
   constraints, practical deployment challenges

## Output Format:
# Practical Feasibility Analysis

## Idea N: [One-sentence Summary]

### Falsifiability & Testability (Score: X/10)
- Testability, Measurability, Observability, Reproducibility

### Experimental Feasibility (Score: X/10)
- Experiment Design, Data Collection, Equipment Availability,
  Ethical Constraints

### Resource Requirements
- Budget: Equipment, Personnel, Materials, Overhead, Total Estimate
- Personnel: Roles, Expertise, Time commitment
- Equipment: Specifications and requirements
- Materials: Details and quantities

### Timeline
- Phases: Name, Duration, Goals, Deliverables
- Key Milestones
- Total Duration, Critical Path

### Risk Management
- Implementation Obstacles
- Mitigation Solutions
- Contingency Plans

### Overall Assessment
- Overall Practical Score: X/10
- Real-World Applications
- Implementation Readiness
```

H.6 Agent 5: Prof. Qwen Ethics (Ethical Review)

```
You are Prof. Qwen Ethics, an expert in scientific ethics and
societal impact assessment.
```

```
## Your Task:
Analyze creative ideas for ethical implications and societal impact.

## Requirements:
1. Research Ethics: Human/animal risks, data privacy, informed
   consent, conflicts of interest, research integrity
2. Societal Impact: Beneficial uses, potential misuse, equity/access,
   long-term consequences, environmental impact
3. Responsibility & Governance: Accountability, regulatory
   compliance, stakeholder engagement, transparency
4. Multi-Dimensional Scoring: 1-10 scale for each dimension plus
   overall score

## Output Format:
# Ethical & Societal Impact Analysis

## Idea N: [One-sentence Summary]

### Research Ethics (Score: X/10)
- Human/Animal Risks, Data Privacy, Informed Consent,
  Conflicts of Interest, Research Integrity

### Societal Impact (Score: X/10)
- Beneficial Applications
- Potential Misuse
- Equity & Access
- Long-term Consequences
- Environmental Impact

### Responsibility & Governance (Score: X/10)
- Accountability, Regulatory Compliance, Stakeholder Engagement,
  Transparency

### Overall Assessment
- Overall Ethics Score: X/10
- Critical Concerns
- Required Safeguards
- Recommendations
```

H.7 Agent 6: Dr. Qwen Leader (Synthesis and Revision)

```
You are Dr. Qwen Leader, the Chief Researcher and Lead Author of
a top-tier scientific paper.

## Your Goal:
```

```
Synthesize all inputs into a publication-ready scientific hypothesis
report that is novel, rigorous, and impactful.

## ABSOLUTE REQUIREMENTS:
1. YOU ARE THE AUTHOR: Write in single, professional, authoritative
   voice. No 'Agent X suggested...'
2. NO TRACES OF PROCESS: Final output must NOT contain process
   markers or metadata
3. SYNTHESIZE: Internalize all inputs and rewrite them
4. INTEGRATE AND SYNTHESIZE: Analyze ALL ideas and synthesize best
   elements from multiple ideas into unified, stronger hypothesis
5. STRICT LATEX FORMATTING:
   - Inline math: $E = mc^2$
   - Display math: $$\sum_{i=1}^n x_i$$
   - Output \sigma, \alpha directly (NOT \\sigma)

## YOUR CORE CAPABILITIES:

### 1. SYNTHESIZE (Initial Report Creation)
- Input: Literature reviews, creative ideas, multi-faceted analysis
- Task: Select BEST idea and write full hypothesis report
- Goal: 'This report must be accepted by Nature/Science/NeurIPS'

### 2. REVISE (Substantive Improvement)
- Input: Current draft + Critical feedback from Critic Crucible
- Task: Make SUBSTANTIVE changes, not cosmetic rewrites
- Revision Principles:
  - NEW CONTENT > REPHRASING: Add equations, algorithms, datasets
  - ADDRESS ROOT CAUSE: If 'lack of novelty', add new mechanisms
  - QUANTIFIABLE IMPROVEMENTS: Every revision adds 3-5 technical elements

### 3. FINALIZE (Quality Assurance)
- Ensure ALL sections present and complete
- Verify LaTeX formatting correct
- Confirm narrative flows: Gap -> Hypothesis -> Solution

## STANDARD REPORT FORMAT:

## Executive Summary
[2-3 sentences with strong Hook]

## Background and Rationale
[Compelling narrative leading to hypothesis]

## Detailed Hypothesis
```

```
[Precise statement, key variables, testable predictions]

### Core Mechanism
[Key components, interactions, LaTeX formulations]

### Technical Innovations
[3-5 specific innovations with novelty explanation]

### Testable Predictions
[3-5 concrete, falsifiable predictions with metrics]

## Supporting Analysis
[Unified argument for plausibility and importance]

## Methodology
[Specific experimental approach, datasets, baselines]

## Expected Outcomes
[Predicted results and implications]

## Limitations & Future Directions
[Thoughtful constraints and actionable next steps]
```

H.8 Agent 7: Critic Crucible (Quality Review)

```
You are Critic Crucible, a Senior Reviewer for a top-tier journal
(e.g., Nature, Science, NeurIPS).

## Your Persona:
- Extremely hard to please. You only accept the top 1%.
- You HATE incremental improvements.
- You LOVE paradigm shifts, rigorous logic, surprising connections.
- Your goal is to REJECT any work that is flawed, derivative, or boring.

## Your Task:
Provide comprehensive peer review with specific, actionable
improvement suggestions in structured Markdown format.

## Requirements:
1. Identify Strengths & Weaknesses: Cite specific technical details
2. Provide Detailed Improvements: Issue, solution, implementation
   steps, expected impact, technical details
3. Assess Missing Elements: Specify what critical components are absent
4. Evaluate Quality Dimensions: Score novelty, rigor, feasibility,
   clarity on 1-10 scale
5. Provide Overall Quality Score: Numeric score 1-10
```

```
## Output Format:

# Critical Analysis of [Topic]

## Overall Assessment
- Quality Score: X/10
- Recommendation: Accept / Revise / Reject
- Reasoning: [Justification. Be direct.]

## Strengths (Why this might be accepted)
- Strength 1, Strength 2, ...

## Weaknesses (Why I want to reject this)
- Weakness 1, Weakness 2, ...

## Critical Concerns (Showstoppers)
[Most important concerns]

## Detailed Improvement Suggestions (How to save this paper)

#### Suggestion N: [Issue Title]
- Issue: [Specific problem]
- Suggestion: [Actionable solution]
- Implementation: [Concrete steps]
- Impact: [Expected improvement]
- Technical Details: [Specific algorithms, methods, references]

[Repeat for 5-8 suggestions if score < 9]

## Missing Elements
- Element 1, Element 2, ...

## Quality Dimensions
- Technical Soundness: X/10
- Novelty Assessment: X/10
- Feasibility: X/10
- Clarity: X/10
```

H.9 Agent 8: Prof. Qwen Editor (Final Polish)

```
You are Prof. Qwen Editor, a distinguished scientific editor at
a top-tier publisher (e.g., Nature Publishing Group).

## Your Goal:
Transform a scientifically sound draft into a compelling, elegant,
```

```
and high-impact manuscript. You elevate the narrative.

## Core Capabilities:

### Mode 1: Content Editor (Structural Improvements)
Use when review indicates:
- Structural issues (reorganize sections)
- Missing content (add baselines, protocols)
- Logic gaps (unclear connections)

Actions:
- Reorganize paragraphs and sections
- Add transition sentences
- Strengthen argumentation
- Expand brief sections

### Mode 2: Language Editor (Polishing)
Use when content is solid but needs refinement:
- Grammar and punctuation errors
- Awkward phrasing or wordiness
- Inconsistent terminology
- Tone inconsistencies

Actions:
- Fix grammatical errors
- Improve sentence flow
- Ensure consistent academic tone
- Perfect LaTeX formatting

## ABSOLUTE REQUIREMENTS:

1. OUTPUT ONLY THE POLISHED DOCUMENT: No editorial analysis,
   scores, or recommendations in final output

2. PRESERVE ALL CONTENT: Do NOT remove sections, equations,
   or citations. Improve, don't delete.

3. MAINTAIN SCIENTIFIC ACCURACY: Don't change technical meaning.
   If unclear, improve explanation.

4. STRICT LATEX FORMATTING:
   - Inline math: $E = mc^2$
   - Display math: $$\sum_{{i=1}}^n x_i$$
   - Output \sigma, \alpha directly
```

```
## Editing Checklist (Complete ALL):
- [ ] All sections preserved
- [ ] Grammar and punctuation correct
- [ ] LaTeX equations render properly
- [ ] Terminology consistent
- [ ] Narrative flows: Gap -> Hypothesis -> Solution
- [ ] Tone authoritative and professional
- [ ] No process markers remain

## REMEMBER:
Your output is the FINAL version that will be saved and evaluated.
Make it publication-ready.
```

I Complete Workflow State Machine Configuration

The multi-agent workflow is defined by a YAML configuration specifying state transitions, agent assignments, and quality gates. Table 12 summarizes the state transition rules.

I.1 Complete Workflow YAML Configuration

```
# Scientific Hypothesis Generation Workflow Configuration
# Based on HypothesisTeam state machine design

workflow:
  name: 'Scientific Hypothesis Generation Workflow'
  description: '8-agent collaborative scientific hypothesis generation'
  version: '2.0'

# Workflow state definitions
states:
  - name: 'INIT'
    display_name: 'Initialization'
    description: 'Initialize team and clear previous results'
    agents: []
    parallel: false
    required: true

  - name: 'LITERATURE'
    display_name: 'Literature Review'
    description: 'Literature review by Scholar Scour'
    agents: ['Scholar Scour']
    parallel: false
    required: true
    timeout: 300  # 5 minutes
```

```
  - name: 'IDEATION'
    display_name: 'Creative Ideation'
    description: 'Hypothesis generation by Idea Igniter'
    agents: ['Idea Igniter']
    parallel: false
    required: true
    timeout: 300

  - name: 'ANALYSIS'
    display_name: 'Parallel Analysis'
    description: 'Parallel review by 3 analysis agents'
    agents: ['Dr. Qwen Technical', 'Dr. Qwen Practical',
             'Prof. Qwen Ethics']
    parallel: true
    required: true
    timeout: 600  # 10 minutes

  - name: 'SYNTHESIS'
    display_name: 'Synthesis Integration'
    description: 'Report writing by Dr. Qwen Leader'
    agents: ['Dr. Qwen Leader']
    parallel: false
    required: true
    timeout: 300

  - name: 'REVIEW'
    display_name: 'Peer Review'
    description: 'Quality control by Critic Crucible'
    agents: ['Critic Crucible']
    parallel: false
    required: true
    timeout: 300
    quality_check: true

  - name: 'REVISION'
    display_name: 'Iterative Revision'
    description: 'Revision by Dr. Qwen Leader'
    agents: ['Dr. Qwen Leader']
    parallel: false
    required: false
    timeout: 300
    iterative: true

  - name: 'POLISH'
```

```
    display_name: 'Final Polishing'
    description: 'Editing by Prof. Qwen Editor'
    agents: ['Prof. Qwen Editor']
    parallel: false
    required: true
    timeout: 300

  - name: 'FINISH'
    display_name: 'Completion'
    description: 'Workflow completion'
    agents: []
    parallel: false
    required: true

# State transition rules
transitions:
  INIT:
    - target: 'LITERATURE'
      condition: 'always'

  LITERATURE:
    - target: 'IDEATION'
      condition: 'success'
    - target: 'FINISH'
      condition: 'failure'

  IDEATION:
    - target: 'ANALYSIS'
      condition: 'success'
    - target: 'FINISH'
      condition: 'failure'

  ANALYSIS:
    - target: 'SYNTHESIS'
      condition: 'success'
    - target: 'FINISH'
      condition: 'failure'

  SYNTHESIS:
    - target: 'REVIEW'
      condition: 'success'
    - target: 'FINISH'
      condition: 'failure'

  REVIEW:
```

```
    - target: 'POLISH'
      condition: 'quality_threshold_met'
    - target: 'REVISION'
      condition: 'needs_improvement'
    - target: 'FINISH'
      condition: 'failure'

  REVISION:
    - target: 'REVIEW'
      condition: 'always'

  POLISH:
    - target: 'FINISH'
      condition: 'always'

# Global configuration
global_config:
  max_retries: 3
  default_timeout: 300
  enable_logging: true
  log_level: 'INFO'

  iteration:
    max_iterations: 3
    quality_threshold: 8.0  # 1-10 scale
    enable_iteration: true
    track_iteration_progress: true

# Error handling
error_handling:
  on_timeout: 'fail_state'
  on_agent_error: 'retry'
  max_state_retries: 2
  retry_delay: 10  # seconds
```

J Complete List of 150 Research Questions

Table 13 presents the statistical distribution of the 150 research questions used in our evaluation.

J.1 Research Question List

Table 14 presents the complete list of 150 research questions.

Table 14: Complete List of 150 Research Questions.

ID	Research Question	Source
1	How can we integrate spatial crime dependencies with temporal dynamics to improve forecasting accuracy?	10.1609/aaai.v36i4.20338
2	How can rules be iteratively refined using newly identified high-priority instances to adapt to evolving data distributions and improve long-term prioritization accuracy?	10.1609/aaai.v33i01.33019440
3	How can pseudo-parallel data generated from structurally informative regions of the unmatched target KG improve the training of a collective relation integration model?	10.18653/v1/2021.acl-long.363
4	How can gesture-enhanced handwriting be effectively combined with contextual insertion mechanics to support efficient deletion and modification of machine-translated text?	10.18653/v1/2020.acl-main.155
5	How can we design a dynamic programming framework that uses ordered representatives to construct locally rainbow paths of bounded detour length?	10.1609/aaai.v38i8.28639
6	How can adversarial text generation be designed to systematically evade deep sequence embedding-based classification models while maintaining semantic coherence and task-specific relevance?	10.1609/aaai.v37i13.26811
7	How can aspect-opinion pairs be effectively represented in a domain-agnostic knowledge graph to capture their contextual sentiment relationships?	10.1609/aaai.v36i11.21653
8	How can a claim verification system effectively integrate and reason over both textual and tabular information in long, hybrid-content financial documents?	10.18653/v1/2024.emnlp-main.818
9	How can hierarchical graph structures be designed to expand concept representations with contextual information from knowledge graphs while preserving structural coherence?	10.1609/aaai.v35i7.16796
10	How can question-answering systems systematically identify and differentiate between presuppositional and implicatural pragmatic inferences in health-related queries?	10.18653/v1/2024.naacl-long.403
11	How can closed-form lower bounds for $\alpha'(N,\alpha)$ be derived to enable fast calibration of NPSS without relying on computationally expensive randomization tests?	10.1609/aaai.v36i4.20339
12	How can fault prediction models be enhanced with data augmentation techniques to improve the accuracy of software bug localization?	10.1609/aaai.v33i01.33019446

Continued on next page

Table 14 – Continued

ID	Research Question	Source
13	How can multiple types of argumentation knowledge graphs be effectively combined to guide the generation of factually consistent and semantically rich arguments?	10.18653/v1/2021.acl-long.366
14	How can a large-scale, domain-specific stance detection dataset for financial M&A events be constructed with high inter-annotator reliability and balanced class distribution?	10.18653/v1/2020.acl-main.157
15	How can a formula be decomposed into a unary function applied to a single parameter and a second function operating on the remaining inputs, enabling recursive isolation of structural components?	10.1609/aaai.v38i8.28641
16	How can a deep learning-based pre-classifier effectively distinguish between normal and abnormal weld seam images to filter out invalid cases before detection?	10.1609/aaai.v37i13.26834
17	How can graph-based link prediction be effectively integrated into cross-domain clustering to improve cluster quality?	10.1609/aaai.v36i11.21654
18	How can a dual-stream architecture with adaptive fusion strategies improve the robustness of multimodal ASR by leveraging both vision hotword streams and traditional ASR outputs?	10.18653/v1/2024.emnlp-main.821
19	How can pre-training strategies be adapted to effectively leverage knowledge graph structures for improving generalization to unseen concept sets in generative commonsense reasoning tasks?	10.1609/aaai.v35i7.16796
20	How can regularization techniques be integrated into annotator embedding learning to prevent overfitting, especially for sparse or low-contribution annotators?	10.18653/v1/2024.naacl-long.407
21	How can human annotation experiments be designed to systematically evaluate the differences between emotions expressed by writers and those perceived by readers, and what implications do these diff...	10.18653/v1/2021.findings-emnlp.219
22	How can a prior over world-models be designed to favor models that assume the external world is static during an episode, thereby preventing an agent from forming plans that require long-term envir...	10.1609/aaai.v34i03.5628
23	How can intra-diversity constraints be formulated to ensure generated solutions remain near-optimal while inter-diversity constraints enforce sufficient differences between solutions?	10.1609/aaai.v34i02.5512

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Table 14 – Continued

ID	Research Question	Source
24	How can the numerical instability of the intersection matrix in skeleton decomposition be mitigated to improve approximation accuracy in self-attention?	10.1609/aaai.v36i1.19962
25	How can adversarial perturbations be optimized to concentrate on dominant frequency regions that are consistently influential across different deep neural networks?	10.1609/aaai.v38i6.28427
26	How can a high-quality textual relation embedding be learned using global co-occurrence statistics between textual and knowledge base relations as supervision?	10.18653/v1/p19-1127
27	How does the scale of transformer language models (e.g., parameter count) influence their ability to process and adapt to atypical animacy, and what does this reveal about their internal representa...	10.18653/v1/2023.emnlp-main.744
28	What is the optimal method for identifying the most relevant evidence sentence in an academic abstract that contains the answer to a given question?	10.1609/aaai.v33i01.33019855
29	How can the ethical risks of NLP crowdsourcing tasks be systematically assessed using the Belmont Report's principles to ensure alignment with human-subjects research ethics?	10.18653/v1/2021.naacl-main.295
30	How can range-null space decomposition be leveraged to enforce strict data consistency in multi-contrast MRI reconstruction while minimizing information loss?	10.1609/aaai.v38i2.27869
31	How do model size, decoding strategies, and prompting with training data affect the novelty of LM-generated text?	10.18653/v1/2024.emnlp-main.800
32	How can the integration of re-anchoring with auxiliary strategies like transfer learning and interactive attention further improve the robustness of implicit discourse relation classifiers?	10.18653/v1/2021.findings-emnlp.110
33	How can spherical coordinate perturbation and pixelwise depth intervals be combined to refine normal and depth estimates in textureless regions?	10.1609/aaai.v38i7.28512
34	How can a single adapter module be trained to adapt a multilingual NMT model to arbitrary new domains and language pairs without requiring domain- or language-specific parameter updates?	10.18653/v1/2022.findings-emnlp.315
35	How can a multi-layer funnel pooling strategy in Conformer encoders achieve progressive frame rate reduction while maintaining sufficient temporal resolution for accurate speech recognition?	10.18653/v1/2024.naacl-long.344

Continued on next page

Table 14 – Continued

ID	Research Question	Source
36	How can an integrated evaluation framework be designed to systematically assess multiple dimensions of metric quality, including correlation, generalization, robustness, and coherence judgment, in ...	10.18653/v1/2021.acl-long.500
37	How can an online agent balance adherence to learned behavioral constraints with the need to maximize reward in a dynamic environment?	10.1609/aaai.v33i01.33013
38	How can federated learning achieve group fairness through dynamic aggregation weight adjustment based on misalignment between local and global fairness metrics?	10.1609/aaai.v37i6.25911
39	Can multilingual transformer models be effectively adapted for cross-lingual acceptability classification tasks, and what are the limitations of this approach for low-resource languages like Italian?	10.18653/v1/2021.findings-emnlp.250
40	How can a domain-specific pre-training dataset for Vietnamese hate speech detection be automatically generated to capture the linguistic diversity and contextual complexity of social media content?	10.18653/v1/2024.findings-acl.355
41	How can external biomedical knowledge graphs be integrated with AMR representations to improve the understanding of domain-specific entities and events?	10.18653/v1/2021.acl-long.489
42	How can a utility function be modified to ensure robustness against (agent $\times$ item)-level differential privacy constraints while preserving fairness guarantees?	10.1609/aaai.v37i5.25721
43	How can spatial and frequency domain information be effectively fused within a diffusion model to improve low-light image restoration quality?	10.1609/aaai.v38i5.28273
44	How can a multi-task training strategy be implemented to jointly optimize hyper-relational KG embedding and contrastive learning objectives without degrading performance?	10.18653/v1/2024.findings-acl.171
45	How can manifold learning be integrated with probabilistic modeling to prevent information loss and improve the accuracy of deformation distribution estimation?	10.1609/aaai.v36i11.21637
46	How can we learn pose-invariant latent representations for single-view 3D reconstruction that are robust to variations in object orientation?	10.1609/aaai.v36i1.19920
47	What is the effectiveness of a RoBERTa-based political stance classifier in identifying biased opinions within generated summaries?	10.48550/arxiv.2402.00322

Continued on next page

Table 14 – Continued

ID	Research Question	Source
48	How can an adversarial training mechanism be designed to prevent the generator from relying on simple heuristics when producing entailment-aware paraphrases?	10.1609/aaai.v36i10.21376
49	What is the impact of incorporating automatically generated affirmative interpretations into the input of pre-trained models for robustness against negation?	10.18653/v1/2024.acl-short.55
50	How can a model-agnostic post-processing framework effectively integrate with diverse summarization models without requiring retraining?	10.18653/v1/2024.findings-acl.59
51	Can feature-based interpretability methods produce significantly different attributions for functionally equivalent neural text classifiers with distinct random initializations?	10.18653/v1/2021.findings-acl.327
52	How can feature importance be leveraged to avoid unnecessary reconstruction of unimportant features during pruning?	10.1609/aaai.v34i07.6720
53	How can few-shot learning strategies be integrated with calibration techniques to further reduce miscalibration in Massively Multilingual Language Models (MMLMs) for languages with limited resources?	10.18653/v1/2022.emnlp-main.290
54	How can the ReviewCritique dataset be utilized to benchmark the effectiveness of both closed-source and open-source LLMs in scientific peer review tasks?	10.18653/v1/2024.emnlp-main.292
55	How can a query-based online search algorithm tolerate up to H erroneous binary responses while efficiently narrowing down the optimal price range?	10.1609/aaai.v36i9.21199
56	How can Contrastive Decoding be adapted to dynamically adjust its strength based on the confidence of the expert model to avoid degrading non-hallucinated translations?	10.18653/v1/2024.eacl-long.155
57	How can channel-wise heterogeneous kernel replacement be designed to adaptively select complementary 1D/2D/3D sub-kernels for each channel in 3D networks?	10.1609/aaai.v35i4.16433
58	How can the joint optimization of syntactic structure and visual-semantic alignment be achieved in an end-to-end unsupervised framework?	10.18653/v1/n19-1180
59	What graph update mechanism can effectively capture both the relational structure of multi-party dialogues and the temporal evolution of emotional states?	10.18653/v1/2022.acl-long.24

Continued on next page

Table 14 – Continued

ID	Research Question	Source
60	How can continuous topic-word distributions be sampled in a differentiable manner to enable end-to-end learning of sentiment-aware topics?	10.18653/v1/2021.eacl-main.199
61	How can a generative adversarial network be adapted to produce high-quality, class-specific embeddings that enhance few-shot learning without memorizing training examples?	10.18653/v1/2022.naacl-main.404
62	How can a shared Attention Residual Gaussian Distribution (ARGD) module enhance the integration of textual information into the posterior path of a variational autoencoder for sign language transla...	10.1609/aaai.v38i17.29937
63	How can data augmentation techniques improve the generalization of Gin Rummy hand evaluation models trained on limited or biased datasets?	10.1609/aaai.v35i17.17843
64	How can spatio-temporal dependencies in agricultural pricing and volume data be jointly modeled using dual temporal convolutional networks to improve prediction accuracy?	10.1609/aaai.v34i08.7039
65	How can a multi-branch mixture density network effectively model the distribution of auction durations by combining game-theoretic predictions with product embeddings?	10.1609/aaai.v38i9.28854
66	How can a multi-slot mechanism iteratively specialize variable representations to capture complex contextual dependencies in mathematical text?	10.18653/v1/2022.findings-acl.76
67	What are the key structural properties of short clauses and low-occurrence variables that the adaptive branching rules exploit to achieve improved performance over prior methods?	10.1609/aaai.v35i5.16479
68	How can a relation classifier be trained to produce confident predictions for individual sentences in an unsupervised setting?	10.18653/v1/p19-1133
69	What evaluation metric can effectively measure the quality of predicted explanations by incorporating human judgments of intuitiveness, even when multiple valid explanations exist?	10.1609/aaai.v36i11.21577
70	How can zero-shot translation capabilities of multilingual MT models be effectively repurposed for grammatical error correction without task-specific adaptation?	10.18653/v1/2024.eacl-long.73
71	How can cell-level layout information be encoded to enable models to recognize semantically coherent units spanning multiple words within a single cell?	10.18653/v1/2021.acl-long.493

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Table 14 – Continued

ID	Research Question	Source
72	How can the outputs of a pattern-exploitation module be integrated with dynamic context augmentation to improve cross-domain generalization in NED?	10.18653/v1/2022.findings-emnlp.506
73	How can dialog systems effectively detect and respond to negative grounding signals (e.g., inconsistencies, errors) to enable collaborative error recovery?	10.18653/v1/2021.eacl-main.41
74	How can a time series prediction model dynamically adapt to the distribution of extreme and normal values while maintaining robustness to imbalanced data?	10.1609/aaai.v37i7.26045
75	How does subfamily-specific pretraining with custom tokenizers enhance model performance for low-resource languages compared to generic multilingual tokenization strategies?	10.18653/v1/2023.findings-eacl.93
76	How can a shared expert mechanism with routing weight normalization be designed to address scale mismatch and improve the effectiveness of sparse upcycling in instruction-tuned code LLMs?	10.18653/v1/2024.acl-long.699
77	How can a semantic latent space be identified and utilized for controllable, accumulative, and composable image editing in transformer-based Flow Matching models?	10.1609/aaai.v38i3.27998
78	Can language models generalize monotonicity-based NPI licensing to unseen environments without explicit training on NPI occurrences?	10.18653/v1/2021.findings-acl.439
79	How can classical optimization techniques be adapted to train quantum classifiers with high accuracy on real-world datasets?	10.1609/aaai.v35i18.17941
80	How can an alternating optimization strategy ensure convergence in online pairwise learning without relying on buffer storage or i.i.d. assumptions?	10.1609/aaai.v38i10.28955
81	How can NLP and ML research better integrate democratic theories to develop frameworks that ensure equitable participation, accountability, and alignment with democratic values in AI systems?	10.18653/v1/2024.emnlp-main.184
82	How does pre-expansion pruning in trie-constrained decoding improve the accuracy and efficiency of parallel sentence retrieval compared to post-expansion pruning?	10.18653/v1/2020.acl-main.152
83	How can transferable component prototypes for individual human-annotated attributes be learned and reused to construct compositional representations for novel classes?	10.1609/aaai.v37i7.26082

Continued on next page

Table 14 – Continued

ID	Research Question	Source
84	How can Chinese characters be effectively transformed into a Latinized stroke sequence representation to leverage subword modeling techniques in NMT?	10.18653/v1/2022.emnlp-main.434
85	How can a contrastive self-training framework with adaptive temperature control improve classification performance by balancing confidence and generalization in pseudo-label learning?	10.18653/v1/2024.eacl-long.39
86	How can adversarial effectiveness in NMT be evaluated without relying on ground-truth reference translations?	10.18653/v1/2021.acl-long.153
87	How can answer spans in paragraphs be leveraged to align answerable and unanswerable question pairs, ensuring both relevance and difficulty?	10.18653/v1/n19-1415
88	How can motion conditions be efficiently mapped to tri-plane feature deformations without using 3D parametric models?	10.1609/aaai.v38i7.28473
89	How can a TableQA benchmark be constructed to systematically evaluate robustness against diverse, human-annotated adversarial perturbations in table headers, content, and questions?	10.18653/v1/2023.acl-long.334
90	How can an alternating optimization algorithm be designed to jointly estimate true labels and train DNN parameters in the presence of instance-dependent label noise?	10.1609/aaai.v35i11.17221
91	How can sequential visual and positional features be adaptively fused to improve robustness in UAV navigation under dynamic environmental conditions?	10.1609/aaai.v38i9.28906
92	How can latent temporal motion information be effectively predicted and encoded in still images for action recognition?	10.1609/aaai.v34i03.5653
93	What architectural design or training strategy enables entity linking models to robustly handle NIL prediction without overfitting to non-NIL examples?	10.18653/v1/2023.findings-acl.690
94	How can conversational patterns from the Natural Conversation Framework be systematically integrated into goal-oriented dialog datasets to reflect naturalistic user behavior?	10.18653/v1/2020.findings-emnlp.358
95	How does the paper design a pre-training task for multi-hop retrievers that enables iterative document retrieval for sub-questions without requiring human annotations?	10.18653/v1/2021.findings-acl.62

Continued on next page

Table 14 – Continued

ID	Research Question	Source
96	How can a model determine the optimal position to insert a simile into a given text to maximize contextual coherence?	10.1609/aaai.v35i16.17691
97	How can we generate logically coherent answer-explanation pairs in VQA-NLE by modeling reasoning steps as a chain-of-thought process?	10.1609/aaai.v38i3.28065
98	How can global property representations be adaptively fused with local sentence features to improve relation prediction?	10.18653/v1/2022.findings-naacl.128
99	How can symptom status recognition be improved by formulating it as a natural language inference task that integrates both dialogue context and external medical knowledge?	10.18653/v1/2023.findings-acl.652
100	How can the trade-off between attack transferability and stealthiness be dynamically controlled during adversarial perturbation generation in the latent space?	10.1609/aaai.v38i4.28147
101	How can human annotation strategies be optimized to effectively evaluate the quality of contrastive explanations in NLP tasks?	10.18653/v1/2023.emnlp-main.427
102	How can hard and soft constraints be effectively combined with pre-trained language models to guide high-quality constrained generation without task-specific training?	10.18653/v1/2020.findings-emnlp.115
103	How can global label hierarchy be effectively encoded into label embeddings using a pre-trained language model without requiring auxiliary graph encoders or intentional fusion of hierarchical and s...	10.18653/v1/2022.emnlp-main.268
104	How can low-performing or data-inefficient clients in federated learning be automatically identified and excluded to improve global model accuracy in autonomous driving simulations?	10.1609/aaai.v37i13.26984
105	How can action and information flow relationships between sentences in procedural texts be inferred without explicit named entity recognition or domain-specific rules?	10.18653/v1/2021.findings-acl.345
106	How can segmentation uncertainty be quantified and visualized through dynamically adjustable predictive regions that reflect user-specified confidence levels?	10.1609/aaai.v37i13.26837
107	How can structured semantic representations for events and arguments be learned from text and visual data in a weakly supervised manner without requiring explicit cross-modal annotations?	10.18653/v1/2020.acl-main.230

Continued on next page

Table 14 – Continued

ID	Research Question	Source
108	How do different architectural designs and pretraining objectives of pretrained language models affect their ability to capture and represent document-level discourse information across multiple la...	10.18653/v1/2021.naacl-main.301
109	Does a text-based hate speech support set outperform a vision-language hate speech support set in few-shot in-context learning for cross-modal hate speech detection?	10.18653/v1/2024.emnlp-main.445
110	How can non-autoregressive translation models be trained to progressively learn bilingual knowledge from fine-grained units (words/phrases) to full-sentence structures?	10.18653/v1/2021.findings-acl.247
111	How can adversarial test cases be generated by systematically replacing answer-relevant spans in evidence with alternative answers while maintaining contextual coherence?	10.18653/v1/2024.findings-naacl.85
112	How can adversarial training be adapted to improve zero-shot cross-lingual transfer by encouraging robustness to embedding misalignments in multilingual representations?	10.18653/v1/2021.emnlp-main.126
113	How can the decoder be modified to interpret intermediate encoder representations in encoder-decoder Transformers?	10.18653/v1/2024.findings-naacl.296
114	How can the integration of auxiliary tasks during the transition phase improve model stability and prevent catastrophic forgetting when shifting from formal to informal data?	10.18653/v1/2020.findings-emnlp.282
115	How can dataset splits be redesigned to mitigate biases from source/time overlap and improve the reliability of evaluation metrics for unreliable news detection?	10.18653/v1/2021.eacl-main.211
116	How can patch-level visual patterns from head classes be effectively transferred to tail classes through self-distillation to alleviate their under-representation?	10.1609/aaai.v38i6.28459
117	How can a branch-and-bound algorithm be designed to iteratively reduce the relaxation gap of the enhanced SDP relaxation and converge to the exact solution?	10.1609/aaai.v37i12.26745
118	How can a value tagger be used to filter noisy or semantically incorrect rewritten utterances in synthesized multi-domain dialogs?	10.18653/v1/2023.emnlp-main.946
119	What methods can enable LLMs to engage in meaningful, multi-turn discussions or debates on abstract topics (e.g., philosophy, ethics) while maintaining coherence, avoiding contradictions, and refle...	10.18653/v1/2023.emnlp-main.146

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Table 14 – Continued

ID	Research Question	Source
120	How can variational inference be adapted to the probabilistic CCA framework to efficiently approximate the posterior distribution of the shared latent variable and view-specific factors in a deep g...	10.1609/aaai.v35i9.16982
121	How can a weakly supervised segmentation model be integrated into a detection framework to achieve pixel-level damage localization using only bounding box annotations?	10.1609/aaai.v34i09.7110
122	How can a dual decoupling framework effectively separate clean and noisy pseudo boxes to improve training stability and performance in semi-supervised object detection?	10.1609/aaai.v36i3.20264
123	How can a funnel-shaped neural network architecture be designed to achieve high compression efficiency while maintaining reconstruction fidelity for spectrum-concentrated data blocks?	10.1609/aaai.v37i4.25602
124	How can instrumental variables be generated under unconditional moment restrictions to eliminate confounding effects in graph representation learning?	10.1609/aaai.v37i6.25925
125	How can combining embeddings from both shallow and deep layers of a transformer model enhance sentence representation quality?	10.18653/v1/2021.findings-emnlp.23
126	How can structured relational knowledge (from hyperlinks) and unstructured factual knowledge (from documents) be dynamically fused during retrieval to improve evidence selection?	10.1609/aaai.v35i15.17568
127	How can the capacity of adapter modules be dynamically adjusted to balance performance gains and computational efficiency across tasks with varying data sizes or complexity?	10.18653/v1/d19-1165
128	How can a multi-hop selection mechanism be designed to iteratively filter the most relevant context utterances for response matching in multi-turn conversations?	10.18653/v1/d19-1011
129	How can symbolic search be combined with neural proof generation to explore diverse and valid proof paths in large search spaces?	10.18653/v1/2022.emnlp-main.7
130	How does the hierarchical organization of discourse units, as defined by Rhetorical Structure Theory, influence the predictability of information density fluctuations?	10.18653/v1/2024.emnlp-main.1047
131	How can relative position representations improve the modeling of long-range dependencies in source code summarization compared to absolute position encodings?	10.18653/v1/2020.acl-main.449

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Table 14 – Continued

ID	Research Question	Source
132	How can graph neural networks be designed to efficiently encode high-order dependency relationships in a direction-aware manner?	10.18653/v1/n19-1237
133	How can conditional gradients be adapted to enforce global convex constraints in deep neural networks while maintaining computational efficiency and convergence guarantees?	10.1609/aaai.v33i01.33014772
134	How can uncertainty-aware contrastive learning be integrated into keypoint detection to improve robustness for texture-less or symmetric objects while maintaining semantic consistency?	10.1609/aaai.v36i5.20482
135	How can local correlation analysis of counterfactual examples be used to disentangle syntactic influences from distributional biases in nationality-related name effects?	10.18653/v1/2024.emnlp-main.34
136	What treebank selection strategy maximizes parsing performance for low-resource languages by combining training data size and linguistic similarity metrics in an automated, heuristic-driven manner?	10.18653/v1/2021.findings-acl.431
137	What automated methods can effectively standardize and annotate complex legal elements (e.g., charges, law articles, penalties) in multi-defendant cases?	10.18653/v1/2024.findings-acl.351
138	How can the teaching order of multiple aligners in a cycle-teaching framework be dynamically optimized to maximize overall performance and minimize error propagation?	10.1609/aaai.v36i4.20348
139	How can a QA system perform compositional reasoning over complex, logically structured documents to identify answers that depend on conjunctions, disjunctions, or exceptions in the text?	10.18653/v1/2022.acl-long.253
140	What is the effectiveness of combining intra-layer self-similarity and inter-layer linear CKA similarity in detecting overfitting of rare and minority group names in neural language models?	10.18653/v1/2021.emnlp-main.41
141	What methods can be used to collect and represent high-quality, diverse annotations from multiple annotators while minimizing bias and ensuring reliability?	10.1609/aaai.v37i6.25840
142	How can the computational efficiency of inference-based filtering be maintained while ensuring high-quality, coherent story outputs?	10.18653/v1/2022.findings-emnlp.520
143	Can the linear optimization problem over the convex polytope $ C_{p^{n+2}} $ be proven #P-hard, and how does this relate to the verification of semivalues for linear thresh...	10.1609/aaai.v33i01.33011869

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Table 14 – Continued

ID	Research Question	Source
144	How can multiple distance metrics be combined into a unified loss function to improve cross-lingual alignment performance across diverse language pairs?	10.18653/v1/2021.findings-acl.276
145	How can source-side information be utilized to improve the retrieval of relevant examples in kNN-MT?	10.18653/v1/2023.emnlp-main.503
146	How can high-quality hard negative samples be generated in the embedding space to improve the discriminative power of contrastive learning for time series?	10.1609/aaai.v38i12.29299
147	How can intra-document structural information be utilized to improve the robustness of contrastive learning models against length-induced semantic drift?	10.18653/v1/2023.emnlp-main.86
148	How can a collaborative framework integrate panoptic-level selection with region-based refinement to achieve efficient and effective active learning for semantic segmentation?	10.1609/aaai.v36i2.20107
149	How can semi-factual data generated using human-identified rationales improve model robustness in few-shot learning without requiring extensive manual intervention?	10.18653/v1/2022.acl-long.481
150	How can a scoring mechanism be designed to prioritize functional correctness by emphasizing recall over precision in evaluating code generation quality?	10.18653/v1/2023.emnlp-main.859

Semantic Unit	Prompt (Extraction Requirements)
Core Problem	The single most important research question the paper focuses on; expressed concisely in 1-2 general, problem-oriented questions (e.g., 'How...?', 'Is it possible to...?') without technical details.
Related Work	Summary of prior research ($\geq$200 words, third-person only) covering existing methods, advantages/disadvantages, research gaps; strictly excludes any reference to the current paper's content; emphasizes trends/limitations in literature.
Preliminary Innovation Analysis	Analyze the proposed framework: list innovative modules (format: 'Module X: [Name] - [Function/novelty]') and framework-level innovations (if any); mandatory before generating Research Questions.
Research Questions (RQ)	2-4 distinct technical questions aligned 1-to-1 with identified innovations; each is a novel/difficult challenge, phrased concisely; non-redundant, quality over quantity.
Simplified Research Question	Keyword-rich, searchable version of the corresponding RQ; retains core technical terms, omits full grammar; maximizes uniqueness for academic search engines.
Solution (for Each RQ)	Detailed step-by-step explanation of how the paper addresses the RQ; includes full procedure, component interactions, definitions of abbreviations/coined terms; technically specific, no vague descriptions.
Simplified Solution	Condensed, keyword-focused version of the corresponding Solution; includes key module names/algorithms; retains high-identifiability technical phrases.
Framework Summary	200-300+ word structured overview of the full method: design motivation, component roles, interactions, key strategies, unique value vs. prior work; uses clear organizational expressions (e.g., 'The framework includes...').

Table 9. Core Semantic Unit Types and Corresponding Extraction Prompts. The table lists 8 primary semantic unit *types*. In practice, each paper is parsed into 21 semantic unit *instances*: including metadata attributes (DOI, title, authors, year, conference, venue, citation count, abstract), multiple research questions (RQ 1–4) and their simplified versions, multiple solutions (SOL 1–4) and their simplified versions, and the framework summary.

Baseline	Key Components	Configuration
Virtual Scientists	Agent Architecture	5 specialized agents (Leader, Researcher, Creative, Critic, Editor) with round-robin discussion
	Knowledge Source	Semantic Scholar API + vector retrieval with top-k=10
	Discussion Rounds	3 rounds of iterative refinement with majority voting
AI Scientist-v2	Search Strategy	Beam search with breadth-first tree expansion, branching factor=3
	Experimentation	Automated code generation and execution in sandboxed environment
	Max Iterations	10 iterations per hypothesis branch, early stopping on convergence
CoI-Agent	Chain Structure	Linear chain-of-ideas with 5 evolutionary steps
	Innovation Mode	Evolutionary mutation + combinatorial fusion of idea pairs
	Feedback Loop	Single-pass generation without iterative revision

Table 10. Baseline Method Configurations.

Agent Name	Role Type	Primary Responsibilities
Scholar Scour	Research Assistant	Literature review, evidence gathering, gap identification
Idea Igniter	Innovator	Creative hypothesis generation, cross-domain synthesis
Dr. Qwen Technical	Technical Reviewer	Technical feasibility, algorithmic complexity assessment
Dr. Qwen Practical	Practical Reviewer	Experimental design, resource estimation, timeline planning
Prof. Qwen Ethics	Ethical Reviewer	Ethical implications, societal impact, risk assessment
Dr. Qwen Leader	Coordinator	Synthesis integration, report writing, revision management
Critic Crucible	Quality Controller	Peer review, quality scoring, improvement suggestions
Prof. Qwen Editor	Editor	Final polishing, formatting, LaTeX compliance

Table 11. Agent Roles and Responsibilities.

Current State	Next State	Transition Condition
INIT	LITERATURE	Always
LITERATURE	IDEATION	Literature review complete ($\geq$10 papers)
IDEATION	ANALYSIS	Hypothesis draft complete
ANALYSIS	SYNTHESIS	All 3 parallel reviewers complete
SYNTHESIS	REVIEW	Synthesis document ready
REVIEW	POLISH	Quality score ≥ 8.0 or iteration ≥ 3
REVIEW	REVISION	Quality score < 8.0 and iteration < 3
REVISION	REVIEW	Revision complete
POLISH	FINISH	Final output formatted

Table 12. State Transition Rules.

Table 13. Distribution of 150 Research Questions.

Category	Count	Percentage
By Conference		
AAAI	63	44.7%
ACL	31	22.0%
EMNLP	31	22.0%
NAACL	9	6.4%
EACL	7	5.0%
By Year		
2019	6	4.3%
2020	14	9.9%
2021	33	23.4%
2022	27	19.1%
2023	23	16.3%
2024	38	27.0%